Engineering Applications of Artificial Intelligence 163 (2026) 112624

Contents lists available at ScienceDirect

Engineering Applications of Artificial Intelligence

journal homepage: www.elsevier.com/locate/engappai

Survey paper

Multimodal fusion in speech emotion recognition: A comprehensive review of methods and technologies

Nhut Minh Nguyen [a], Thanh Trung Nguyen [a], Phuong-Nam Tran [b], Chee Peng Lim [c], Nhat Truong Pham [d], Duc Ngoc Minh Dang [a,*]

[a] AiTA Lab, Department of Computing Fundamentals, FPT University, D1 Street, Saigon Hi-tech Park, Tang Nhon Phu Ward, Ho Chi Minh City, 71216, Viet Nam
[b] Department of Artificial Intelligence, Kyung Hee University, Yongin, 446-701, Republic of Korea
[c] Department of Computing Technologies, Swinburne University of Technology, Victoria, 3122, Australia
[d] Department of Integrative Biotechnology, College of Biotechnology and Bioengineering, Sungkyunkwan University, Suwon, 16419, Gyeonggi-do, Republic of Korea

ARTICLE INFO

Dataset link: https://github.com/aita-lab/awesome-multimodal-fusion-emotion-recognition/

Keywords:
Attention mechanisms
Deep learning
Feature fusion
Multimodal analysis
Speech emotion recognition

ABSTRACT

Speech emotion recognition (SER) plays a crucial role in human–computer interaction, enhancing numerous applications such as virtual assistants, healthcare monitoring, and customer support by identifying and interpreting emotions conveyed through spoken language. While unimodal SER systems demonstrate notable simplicity and computational efficiency, excelling in extracting critical features like vocal prosody and linguistic content, there is a pressing need to improve their performance in challenging conditions, such as noisy environments and the handling of ambiguous expressions or incomplete information. These challenges underscore the necessity of transitioning to multimodal approaches, which integrate complementary data sources to achieve more robust and accurate emotion detection. With advancements in artificial intelligence, especially in neural networks and deep learning, many studies have employed advanced deep learning and feature fusion techniques to enhance SER performance. This review synthesizes a comprehensive collection of publications from 2020 to 2024, exploring prominent multimodal fusion strategies, including early fusion, late fusion, deep fusion, and hybrid fusion methods, while also examining data representation, data translation, attention mechanisms, and graph-based fusion technologies. We assess the effectiveness of various fusion techniques across standard SER datasets, highlighting their performance in diverse tasks and addressing challenges related to data alignment, noise management, and computational demands. Furthermore, we highlight real-world applications of multimodal SER and provide critical research challenges that must be addressed for practical deployment, offering insights into optimal fusion strategies and guiding future developments in multimodal SER.

1. Introduction

Emotions play a crucial role in human interaction, influencing how individuals communicate and understand one another in daily activities. Humans express emotions through various channels, including gestures, facial expressions, body posture, and, most significantly, speech (Manalu and Rifai, 2024). Additionally, physiological markers such as heart rate, blood pressure, muscle activity, and skin conductivity provide further insights into emotional states. Among these indicators, speech is one of the most direct and accessible ways to gauge emotion, making it a primary research focus in emotion recognition (Khare et al., 2024; Nguyen et al., 2024a). Identifying emotions from speech involves examining three key aspects: the content of the spoken words, how the speech is conveyed, and the speaker's characteristics, such as gender and voice quality (Gangamohan et al., 2016). Understanding these factors provides a comprehensive view of a speaker's emotional state and drives significant advancements in speech-based emotion recognition technology.

Speech emotion recognition (SER) is a crucial field of study within human–computer interaction, focusing on accurately identifying and interpreting emotions conveyed in spoken language. Applications of SER are vast and continually expanding, spanning multiple domains. In affective computing and human–computer interaction, SER processes acoustic features to detect emotional states, enabling virtual assistants and dialogue systems to respond empathetically and adjust their interaction style in real time (Górriz et al., 2023). Virtual assistants and emotion-aware dialogue systems can use SER to detect user frustration and shift to a more supportive tone (Zheng et al., 2025; Ma et al., 2024; Fei et al., 2024). In healthcare, SER analyzes voice pitch, intensity,

* Correspondence to: AiTA Lab, Department of Computing Fundamentals, FPT University, Ho Chi Minh Campus, Vietnam.
E-mail address: ducdnm2@fe.edu.vn (D.N.M. Dang).

https://doi.org/10.1016/j.engappai.2025.112624
Received 3 December 2024; Received in revised form 31 July 2025; Accepted 2 October 2025
Available online 21 October 2025
0952-1976/

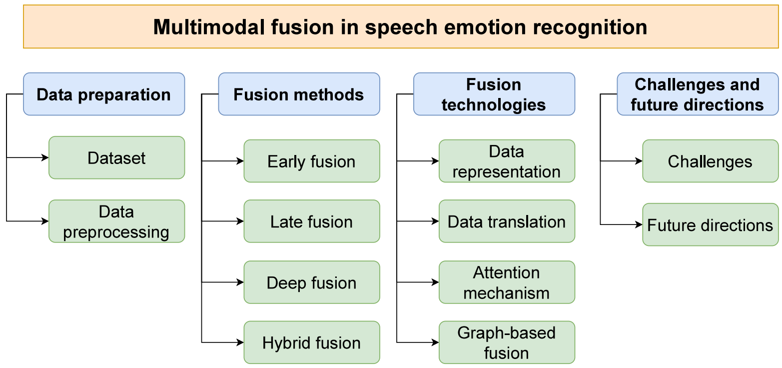


Fig. 1. Survey of multimodal fusion in SER.

and speech rhythm from vocal biomarkers to detect stress and signs of mental health conditions, facilitating early intervention and enabling personalized treatment plans (Wang et al., 2025; Othmani et al., 2021). In education, emotion-aware learning platforms leverage SER to monitor student engagement in real-time, identifying emotional cues like confusion or boredom and triggering adaptive content delivery for personalized learning experiences (Vyakaranam et al., 2024). Moreover, in customer service, SER is integrated into intelligent support systems to detect user frustration or dissatisfaction from speech signals, enabling timely interventions, dynamic support routing, and improved overall service efficiency.

While unimodal SER offers advantages such as reduced computational complexity (Shome and Etemad, 2024) and effective analysis of specific acoustic features like prosody and intonation (Tomar et al., 2023), it has notable limitations. However, unimodal SER has drawbacks; it is vulnerable to background noise, poor recording quality, and variations in speaking styles (George and Ilyas, 2024). Unimodal systems may misinterpret emotions as they rely solely on a single source of signals for emotional interpretation (Hallmen et al., 2024). This limitation often results in decreased accuracy, especially in dynamic environments where multimodal data provides complementary cues for a better understanding (Poria et al., 2017). Additionally, emotions expressed in speech can be ambiguous, and their conveyance can be adversely affected by microphone quality, compression, or transmission noise, particularly in digital communication (Ezzameli and Mahersia, 2023). Recognizing the limitations of unimodal SER approaches, this study focuses on employing multimodal fusion for SER.

Multimodal fusion offers considerable advantages across various fields by enhancing the integration of diverse data types, ultimately improving accuracy and effectiveness. Particularly in SER, a multimodal fusion combining acoustic, visual, and textual features comprehensively represents emotional states. This approach effectively addresses the limitations often encountered with unimodal methods. Nevertheless, integrating multiple data modalities presents significant challenges, including synchronization, data alignment, noise management, and heightened computational demands. These complexities have driven extensive research into robust multimodal fusion methods to enhance SER accuracy in real-world applications. This survey thoroughly reviews leading multimodal fusion techniques utilized in SER, encompassing early, late, deep, and hybrid fusion methods, along with recent advancements in attention mechanisms and graph-based fusion approaches. We systematically compare performance across widely recognized SER datasets to assess the effectiveness of various fusion methods. Moreover, we elaborate on critical challenges and propose future research directions to propel the evolution of multimodal SER systems. The overview of this survey is visualized in Fig. 1.

In this survey, the main contributions are as follows:

- Comprehensive analysis of fusion methods: This survey analyzes various fusion methods applied to SER. It explores early, late, deep, and hybrid fusion methods, highlighting their benefits and limitations in multimodal SER applications.
- Comprehensive review of fusion technologies: We review cutting-edge fusion models, including data alignment, data representations, data translation techniques, attention mechanisms, graph-based fusion, and other advanced multimodal fusion models, to demonstrate how each contributes to enhanced SER accuracy.
- Comparative performance analysis: This survey conducts a detailed performance comparison of existing fusion methods across different datasets. By analyzing weighted accuracy (WA) and unweighted accuracy (UA) metrics, we assess the effectiveness of each method, providing insights into optimal fusion techniques for various SER tasks.
- Discussion of challenges and future research directions: We identify and discuss the primary challenges in multimodal fusion for SER, including data alignment, computational demands, and handling modality-specific noise. Furthermore, we present future research directions to address these challenges and enhance multimodal fusion efficacy in SER.

The remainder of this paper is structured as follows: Section 2 aims to thoroughly analyze and identify appropriate datasets, fusion methods, and network architectures for SER. Section 3 highlights SER through multimodal fusion, presenting the growth in integrating diverse data sources to enhance recognition capabilities. Section 4 describes the datasets and feature extraction for speech and linguistic data in multimodal fusion in SER. Section 5 categorizes feature fusion approaches into early, late, deep, and hybrid fusion, analyzing their benefits and limitations. Section 6 addresses data alignment, data representations, data translation techniques, attention mechanisms, and graph-based fusion vital for effective multimodal fusion. In Section 7, we highlight effective SER models, address the research questions, identify gaps in fusion methods, and suggest future research directions. The concluding remarks of this review are summarized in Section 8.

2. Research objective

2.1. Research questions (RQ)

This survey explores research on multimodal fusion in SER, focusing on datasets, fusion methods, and network architectures. It reviews datasets, analyzes fusion techniques, and evaluates architectures for improved accuracy and interpretability. The study also addresses real-time challenges like data alignment and computational needs, with research questions outlined in Table 1.

2.2. Search process

The study selection process in this paper was executed systematically. We systematically searched prominent databases following the

Table 1
Key research questions for advancing multimodal fusion in SER.

Question	Purpose
RQ 1: What publicly accessible multimodal datasets are commonly used for multimodal fusion in SER study?	To identify and evaluate available datasets that include multiple modalities for SER, aiding researchers in selecting datasets for comprehensive studies. To understand which features in each modality contribute most to emotion recognition accuracy, guiding feature selection for improved model performance.
RQ 2: Which fusion methods and techniques most effectively integrate multimodal data for SER, and how do they impact performance?	To analyze the effectiveness of various fusion methods in combining multimodal data, identifying which techniques enhance emotion recognition in SER models.
RQ 3: How do fusion techniques enhance the accuracy and interpretability of multimodal SER models?	To assess the role of fusion techniques in SER, focusing on how different network architectures and attention mechanisms integrate multimodal features. This evaluation examines their impact on capturing emotional cues, improving model performance, and enhancing interpretability.
RQ 4: What are the main challenges associated with multimodal fusion in SER, and how can they be addressed?	To examine challenges such as data alignment, computational demands, and handling noise across modalities in SER applications and to suggest possible solutions.

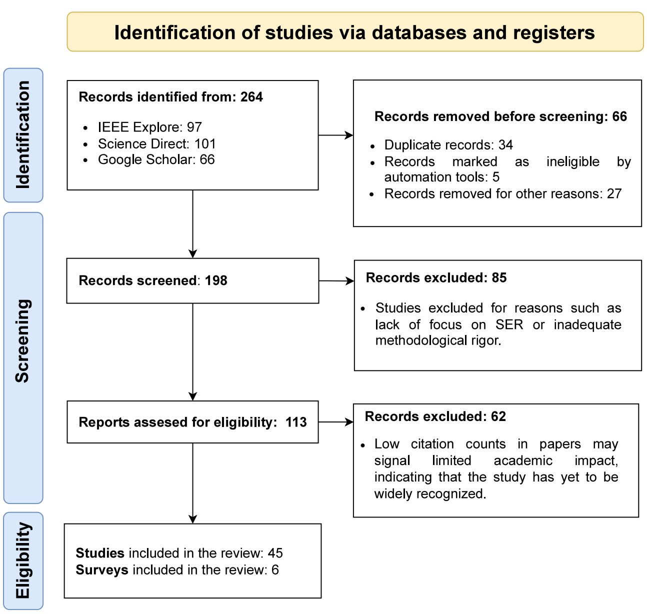


Fig. 2. A PRISMA-based screening and selection of studies in SER.

preferred reporting items for systematic reviews and meta-analyses (PRISMA) (Tugwell and Tovey, 2021) flow diagram, as shown in Fig. 2. The following is outlined below:

- **Search strategy:** We employed precise search keywords and effective boolean operators (AND, OR) to enhance the quality of our results.
- **Digital databases:** We comprehensively searched the IEEE Xplore, ScienceDirect, and Google Scholar databases.
- **Inclusion and exclusion criteria:** We included studies focusing on multimodal fusion in SER that use established datasets and explore various fusion methods. We excluded studies that lack a clear SER focus, address only unimodal approaches, demonstrate methodological weaknesses, and are of low quality based on paper organization, research goals, and paper citation counts.
- **Screening outcome:** In this stage, we remove duplicates and conduct a quality check to include only the best studies addressing the research questions.

2.2.1. *Search strategy*

The criteria for selecting the search keywords were as follows:

- The new terms were derived from articles and published papers.
- Boolean operators (AND, OR) were used to narrow the search results.

The keywords used for the search are as follows:

- "Speech emotion recognition" OR "Voice-based emotion recognition" AND "Survey of speech emotion recognition" OR "Survey of Feature Fusion".
- "Emotional speech database" OR "Speech emotion recognition with multimodal" AND "Speech emotion recognition based feature fusion".
- "Fusion methods and fusion technology" OR "Feature fusion" OR "Feature technology" AND "Speech emotion recognition with multimodal".

2.2.2. *Inclusion criteria*

Studies were included if they met several specific criteria: they focused on multimodal fusion techniques within SER, used established datasets, examined various fusion methods, presented performance at or near the state-of-the-art, and provided insights into network architectures for emotion recognition. Studies that addressed real-time challenges, data alignment, or using graph-based fusion in SER were also prioritized.

2.2.3. *Exclusion criteria*

Papers will be excluded based on several criteria to ensure a focused and rigorous review. Firstly, any paper that lacks a specific focus on SER will be excluded from consideration. Additionally, we will eliminate studies that do not address multimodal fusion methods, as our interest lies in integrating various data modalities. This means that papers concentrating exclusively on unimodal approaches will not be included. Furthermore, we will avoid repeated studies to maintain the originality of our findings. Lastly, we will ensure that the methodologies of the included papers are robust, excluding those with weaknesses such as insufficient data or unclear evaluation metrics. By adhering to these guidelines, we aim to curate a high-quality selection of literature for our analysis.

Table 2
Comparison of the proposed review with the existing reviews in multimodal SER (✓ Include, ✗ Not include).

Review paper	Data preparation		Fusion method				Fusion technologies			
	Dataset	Feature extraction	Early fusion	Late fusion	Deep fusion	Hybrid fusion	Data representation	Data translation	Attention mechanism	Graph-based fusion
Jiang et al. (2020)	✓	✓	✓	✓	✗	✗	✗	✗	✗	✗
Atmaja et al. (2022)	✓	✓	✓	✓	✗	✗	✗	✗	✗	✗
Pan et al. (2023)	✓	✓	✓	✓	✗	✓	✗	✗	✗	✗
Kakuba et al. (2023)	✓	✓	✓	✓	✗	✓	✗	✗	✓	✗
Jiao et al. (2024)	✓	✓	✓	✓	✓	✓	✓	✓	✗	✗
Khan et al. (2024b)	✓	✗	✓	✓	✓	✓	✗	✗	✓	✗
This review	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

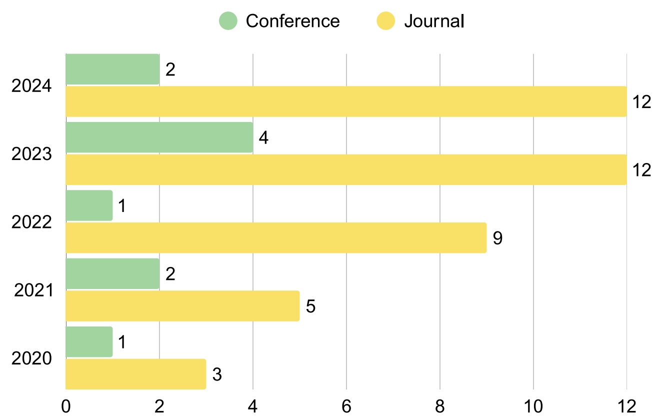


Fig. 3. Publication distribution used in this study.

2.2.4. Screening outcome

After removing duplicates, 198 studies were retained from an initial pool of 264 records. Following a meticulous screening process, we assessed 113 studies, including 45 high-quality studies and six comprehensive surveys in our review. Fig. 3 shows the publication distribution used in our survey, spanning from 2020 to 2024, with 19.61% from conferences/proceedings and 80.39% from journal publications.

3. Related surveys

From 2020 to 2024, SER has seen significant advancements through multimodal fusion methods, integrating data from diverse modalities to improve emotion recognition accuracy. The evolution of fusion methods has been instrumental in enhancing the capabilities of SER systems. A comparison of the proposed review with other existing reviews is presented in Table 2.

Jiang et al. (2020) reviewed the development and challenges of data-driven multimodal SER, emphasizing real-time mental health monitoring systems, available multimodal datasets, feature extraction methods, and fusion methods. The review identifies critical scientific problems and suggests future research directions, advocating for improved fusion techniques to overcome the limitations of unimodal emotion recognition. While the paper focuses on applying electroencephalogram, speech, and text in multimodal fusion, it does not delve deeply into fusion methods, mentioning only two types: early and late fusion. Additionally, it does not discuss the methodologies used for these fusion types.

Atmaja et al. (2022) explored integrating acoustic and linguistic information to enhance the accuracy of emotion detection in speech. Traditional SER approaches rely solely on acoustic features like pitch and intonation, but these methods face performance limitations. The survey highlights that fusing acoustic data with linguistic content derived from automatic speech recognition improves SER outcomes. It reviews various datasets, features, and classifiers used in bimodal SER. The findings show that bimodal systems outperform unimodal ones, especially with deep learning (DL) models and attention mechanisms. However, this review also discusses two popular fusion approaches: early and late fusion. The review only mentions the four most used classifiers in SER in the classifier model, namely, support vector machine (SVM), multi-layer perceptron (MLP), convolutional neural networks (CNNs), and long short-term memory (LSTM) models.

Pan et al. (2023) provided a comprehensive review of multimodal SER, focusing on datasets, pre-processing techniques, feature extraction, and fusion methods. It discusses how machines can recognize human emotions by integrating multiple modalities and emulating human emotional perception. Key aspects include using discrete and dimensional emotion models, detailed pre-processing methods for each modality, and various feature extraction approaches, from handcrafted to DL-based techniques. It also explores three types of fusion methods to improve the accuracy of emotion recognition models. Advanced techniques such as self-supervised learning, graph neural networks (GNNs), and transformers are poised to enhance multimodal SER significantly. These methods effectively capture complex dependencies across diverse data types, making them particularly powerful in low-resource scenarios. The review concludes by identifying challenges and highlighting future research directions.

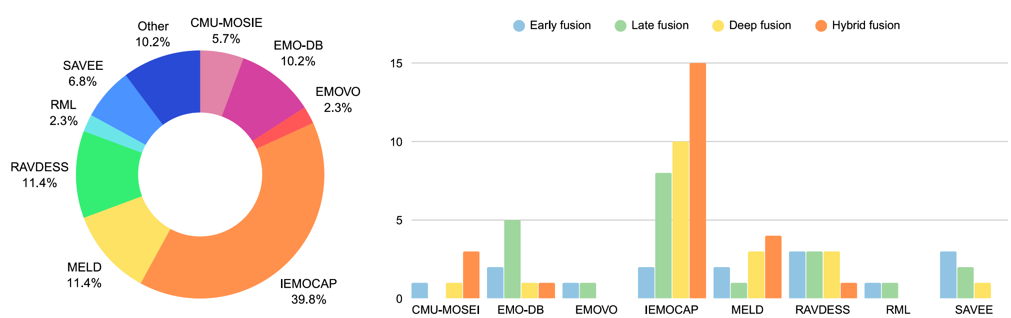
Kakuba et al. (2023) focused on developments in DL for bimodal SER, particularly the use of attention mechanisms and fusion methods integrating acoustic and linguistic data. The review covers innovative techniques like multi-head attention (MHA) mechanisms at various fusion levels, essential for capturing intricate intra- and inter-modality features. Datasets used for multimodal fusion in SER are discussed, but only less popular ones are covered. The author also proposes a multi-technique model called the deep learning-based multi-learning model for emotion recognition (DBMER). A model DBMER that combines CNNs, recurrent neural networks (RNNs), and attention mechanisms to address SER's practical challenges is presented. This model offers a robust and comprehensive review of DL's potential for SER in real-world environments, instilling hope for the future of SER research.

Jiao et al. (2024) provided a comprehensive overview of multimodal fusion methods. It leverages the complementary characteristics of multiple data streams to enhance accuracy in complex environments. The review categorizes multimodal fusion into four essential methods: early fusion, deep fusion, late fusion, and hybrid fusion. The survey also highlights three prominent fusion technologies that improve data fusion effectiveness while discussing the challenges of data heterogeneity and redundancy between modalities. However, it does not cover graph-based fusion methods, which effectively represent relationships in multimodal data. It also lacks a discussion on the importance of data pre-processing and feature extraction. Finally, it suggests future research directions and emphasizes preserving useful information while minimizing noise during the fusion process.

Khan et al. (2024b) comprehensively analyzed multimodal fusion methods used in tourism applications. It explores diverse fusion methods like early, late, hybrid, and hierarchical methods and recent DL models. The survey highlights the importance of integrating data from various sources, including social media, to understand tourism ecosystems better. It also discusses benchmark datasets, federated learning approaches, and the challenges faced in tourism applications, offering insights into future research directions.

Table 3
Datasets of emotion recognition.

Dataset	Subject	Size	Modality	Type	Emotion description
CMU-MOSEI	1,000	Available 23,000 recordings of audiovisual data	Audio, video, and text	Spontaneous emotions in wilds	Sadness, happiness, fear, anger, surprise, and disgust
EMO-DB	111	More than 500 utterances	Audio and video	Spontaneous emotions in labs	Anger, boredom, disgust, fear, happiness, sadness, and neutrality
EMOVO	6	Available 500 audio files.	Audio.	Acted emotions in labs.	Disgust, fear, anger, joy, surprise, sadness and neutrality
IEMOCAP	10	Approximately 12 hours of audiovisual data	Audio, video, and text	Spontaneous emotions in labs	Anger, happiness, excitement, sadness, frustration, fear, surprise, and neutrality
MELD	304	More than 1,400 dialogues and 13,000 utterances	Audio, video, and text	Spontaneous emotions in wilds	Anger, disgust, sadness, joy, neutrality, surprise, and fear
RAVDESS	24	Contains 7,356 files (total size: 24.8 GB)	Audio and video	Spontaneous emotions in labs	Neutrality, calmness, happiness, sadness, anger, fear, disgust, and surprise
RML	8	Available 720 utterances	Audio and video	Acted emotions in labs	Anger, disgust, fear, happiness, sadness and surprise
SAVEE	4	Available 480 utterances	Audio and video	Spontaneous emotions in labs	Anger, disgust, fear, happiness, sadness, surprise, and neutrality



(a) Dataset distribution by number of samples and papers cited.

(b) Dataset distribution across fusion methods.

Fig. 4. Fusion method performance and dataset distribution in SER.

4. Data preparation for SER

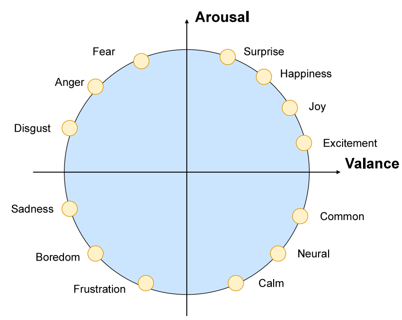
4.1. Datasets

Various datasets are crucial for developing and evaluating different fusion methods in multimodal SER tasks. These datasets include both unimodal and multimodal samples, providing diverse emotional expressions and multimodal data for research. They enable researchers to effectively investigate how to combine information from multiple sources to improve recognition accuracy. Table 3 highlights several key datasets commonly used in SER, providing a foundation for implementing various fusion methods. The efficacy of these fusion methods is depicted in Fig. 4, demonstrating the impact of different techniques on emotion recognition accuracy across various datasets. This visual representation underscores the importance of selecting suitable datasets and fusion methods to achieve enhanced SER results, thereby improving the efficacy of emotion recognition systems across various applications.

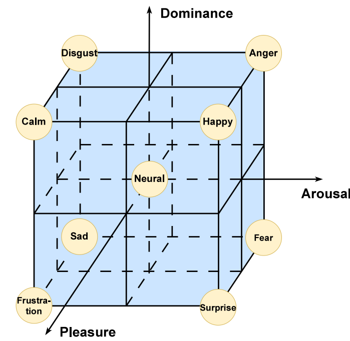
CMU-MOSEI (Zadeh et al., 2020) is the largest dataset for multimodal sentiment analysis and emotion recognition. It contains over 23,500 sentence utterance videos from more than 1,000 different speakers on YouTube, ensuring gender balance. The utterances are randomly selected from various topics and monologue videos. Additionally, all videos are transcribed and correctly punctuated, making CMU-MOSEI a comprehensive resource for studying sentiment and emotion across diverse multimodal contexts.

EMO-DB (Burkhardt et al., 2005) was developed as part of the DFG-funded research project SE462/3-1 conducted between 1997 and 1999. Actors in the Department of Technical Acoustics at the Technical University of Berlin recorded emotive utterances in the anechoic chamber. This dataset is composed of these recordings. Prof. Dr. W. Sendlmeier of the Institute of Speech and Communication oversaw the initiative. Key contributors included Felix Burkhardt, Miriam Kienast, Astrid Paeschke, and Benjamin Weiss. EMO-DB features a spectrum of emotional expressions, providing a valuable resource for research in emotional speech recognition and analysis.

EMOVO (Costantini et al., 2014) was the first emotional database specifically designed for the Italian language. It includes recordings from six actors performing 14 sentences that simulate six emotional and neutral states. These emotions represent the well-known Big Six in emotional speech literature. The recordings were made using professional



(a) VA dimension emotion model (Lo et al., 2018).



(b) PAD dimension emotion model (Mehrabian, 1996).

Fig. 5. Comparison of valence-arousal and pleasure-arousal-dominance emotion models.

equipment at Fondazione Ugo Bordoni Laboratory. The corpus also features a subjective validation test involving emotion discrimination tasks conducted by two groups of 24 listeners, confirming its reliability for studying emotional expression in Italian speech.

IEMOCAP (Busso et al., 2008), a multimodal and multi-speaker resource collected at the SAIL Lab at the University of Southern California in 2008, is one of the most significant datasets in this domain. Comprising approximately 12 h of audiovisual data, it includes video, speech, motion capture of facial expressions, and text transcriptions. The richness of its multimodal data makes it suitable for evaluating various feature fusion methods, including early, late, deep, and hybrid fusion techniques.

MELD (Poria et al., 2018) extends the original EmotionLines dataset by incorporating audio and visual modalities alongside text. The Friends TV series is the source of over 1,400 dialogues and 13,000 utterances in the MELD, with multiple speakers participating in the dialogues. Seven emotions are assigned to each utterance, including anger, disgust, sadness, joy, neutral, surprise, and fear. In addition, MELD offers sentiment annotations (positive, negative, and neutral) for each utterance, thereby establishing it as a comprehensive multimodal resource for sentiment analysis and emotion.

RAVDESS (Livingstone and Russo, 2018) is a validated dataset containing emotional speech and song, with balanced gender representation from 24 professional actors. The actors vocalize lexically matched statements in a neutral North American accent. The speech covers calm, happy, sad, angry, fearful, surprised, and disgusted emotions, while the song includes calm, happy, sad, angry, and frightening emotions. Each expression is presented at two levels: emotional intensity and neutral expression. High emotional validity and test-retest inter-rater reliability were reported, making RAVDESS a highly reliable resource for emotion research.

RML (Wang and Guan, 2008) is a multimodal collection of video samples from eight subjects speaking six languages: English, Mandarin, Urdu, Punjabi, Persian, and Italian, with different accents of English and Mandarin included. For diversity, some subjects have facial hair. Video samples were chosen based on listening tests by at least two non-native speakers to ensure accurate emotional expression. A sample was only included if all testing participants correctly perceived the intended emotion. The recordings were made at a sampling rate of 22,050 Hz and a frame rate of 30 frames per second (fps), with 16-bit digitization on a single channel, offering a rich resource for emotion recognition research across multiple modalities.

SAVEE (Jackson and Haq, 2014) is a validated multimodal dataset containing emotional speech and song recordings. It features a balanced representation of genders, with contributions from 24 professional actors who vocalize lexically matched statements in a neutral North American accent. The speech components encompass a range of emotional expressions, including calm, happy, sad, angry, fearful, surprised, and disgusted, while the song segments include calm, happy, sad, angry, and frightened emotions.

Dimensional emotion models provide a framework for SER datasets by representing emotions as points in a continuous space rather than discrete categories. The two primary models are the valence-arousal (VA) model (Lo et al., 2018), which measures emotions based on pleasantness (valence) and intensity (arousal), and the pleasure-arousal-dominance (PAD) model (Mehrabian, 1996), which adds a control dimension (dominance) to capture the speaker's sense of influence over their emotional state as illustrated in Fig. 5. The VA model represents emotions on a two-dimensional plane: valence captures positivity (right for positive emotions like happiness) and negativity (left for negative emotions like sadness), while arousal indicates intensity (top for high-energy emotions like fear, bottom for low-energy emotions like calm). The PAD model adds a third dimension, dominance, which measures feelings of control. Emotions such as anger and calm are high in dominance, suggesting power, whereas fear and sadness are low, reflecting helplessness. This model offers a broader understanding of emotions by integrating pleasure, arousal, and dominance. These models are particularly suitable for SER applications because they align well with the continuous nature of speech signals and acoustic features, enable more nuanced emotion tracking, and often demonstrate better cross-cultural applicability than discrete labels. Additionally, their continuous representation supports regression-based machine-learning approaches. It allows for more precise mapping between acoustic parameters and emotional states while maintaining the flexibility to convert between dimensional and discrete representations when needed.

As visualized in Fig. 4, the widespread use of datasets like IEMOCAP, RAVDESS, and MELD highlights their significance in advancing SER research. The potential benefits of employing multimodal fusion methods to improve recognition accuracy across various emotional states are evident. Future research will delve deeper into integrating multiple datasets and fusion methods to enhance the robustness and effectiveness of SER systems.

4.2. Feature extractions

The quality and informativeness of input features play a critical role in the performance of bimodal SER systems. Accurate emotion prediction hinges on selecting emotionally expressive features relevant to the intended emotional outcomes. The most effective features for SER can be derived from acoustic and linguistic domains, each contributing unique insights into emotional cues. These features are typically categorized based on the approach used for their extraction:

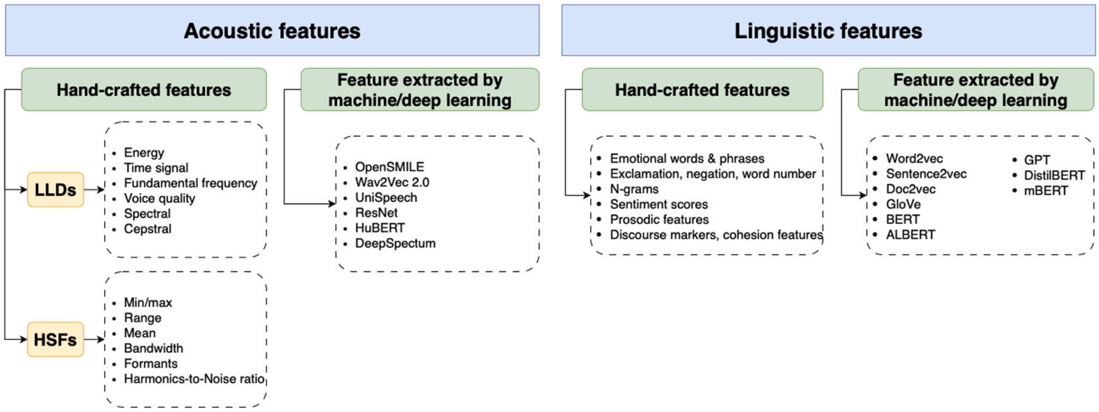


Fig. 6. Classification of acoustic and linguistic features for SER.

some are manually formulated based on established physical models and domain expertise, while others are automatically derived through data-driven learning processes, often using deep learning (DL) models. This classification helps to understand the extraction's underlying methodology and its impact on SER performance. We present a visualization of both acoustic and linguistic features extracted from the datasets in Fig. 6. This figure highlights the complementary nature of these modalities, providing insights into their role in capturing nuanced emotional expressions.

4.2.1. Acoustic features

Acoustic feature extraction plays a crucial role in capturing the relevant characteristics of audio signals in SER and related tasks. Feature extraction methods can be divided into hand-crafted features and features extracted using machine learning (ML) or DL techniques. Hand-crafted features, such as low-level descriptors (LLDs) and high-level statistical functions (HSFs) (Madanian et al., 2023), are comprehensive in their coverage of audio signal properties. LLDs, for instance, encompass essential audio characteristics. These features often describe the signal's temporal, spectral, and tonal qualities (Umapathy et al., 2007). On the other hand, HSFs, derived from LLDs, capture statistical summaries over time. This comprehensive coverage provides a detailed understanding of the audio data often used to assess voice characteristics (Schuller, 2013).

Features extracted by ML or DL models automate the feature extraction, often producing more abstract and complex audio signal representations. Techniques are used to directly learn hierarchical latent features from raw audio or spectrograms. For instance, models like Wav2Vec 2.0 (Baevski et al., 2020) and HuBERT (Hsu et al., 2021) use unsupervised learning to capture rich representations of speech, while ResNet (Han et al., 2021) and DeepSpectrum (Zhao et al., 2018) typically process spectrograms with CNNs. DL-based approaches often outperform traditional hand-crafted methods by capturing more nuanced and high-dimensional features that are challenging to define manually.

4.2.2. Linguistic features

Linguistic features are crucial for analyzing text in emotion detection, sentiment analysis, and various natural language processing (NLP) applications. These features can be broadly categorized into hand-crafted features and those extracted by ML or DL techniques. Hand-crafted linguistic features are manually defined and focus on specific, well-understood aspects of language. Examples include identifying emotional words and phrases, detecting elements like exclamations, negations, and word count, and analyzing n-grams to capture common word patterns. Sentiment scores measure word or phrase positivity, negativity, or neutrality, while prosodic features capture rhythm, stress, and intonation. Additionally, discourse markers and cohesion features highlight connectors and structures that reveal the organization of ideas within the text.

ML- and DL-based approaches often capture high-dimensional representations that reflect deeper semantic and syntactic nuances. Models like Word2Vec (Khatua et al., 2019), Sentence2Vec (Zhang et al., 2018), and Doc2Vec (Chen et al., 2016) transform words, sentences, or documents into vector embeddings that preserve semantic relationships. GloVe (Pennington et al., 2014), another popular embedding method, captures global word co-occurrence statistics for nuanced word representations. Advanced transformer-based models provide contextualized embeddings that adapt word meanings based on surrounding text, making them highly effective for various NLP tasks. Generative models like generative pre-trained transformers (GPT) (Sousa et al., 2023) also capture sequential dependencies and contextual information across the text.

In contrast to acoustic features, there is a growing tendency in linguistic feature extraction to shift from traditional hand-crafted features to DL-based approaches. While applicable, hand-crafted linguistic features may not capture the intricate semantic and syntactic relationships that DL models can extract, particularly for complex NLP tasks. The choice of linguistic features often depends on the complexity of the task and the available data size. By leveraging hand-crafted and DL-extracted linguistic features, researchers can build robust models that comprehensively capture language structure, meaning, and emotional content, enhancing performance in NLP applications.

5. Review of feature fusion for SER

The multimodal fusion in SER starts with input data from multiple modalities, each undergoing independent feature extraction. These extracted features are then passed to a fusion mechanism that integrates the multimodal information into a single vector. This unified representation is fed into a final learning model, which processes the fused features and generates the output. Fig. 7 visualizes the flow of feature fusion in multimodal SER. The fusion of multimodal into a single vector can be implemented through various strategies. Concatenation (Zheng et al., 2020) directly combines extracted features from different modalities into a single feature vector. Feature selection (Li et al., 2017) retains only the most relevant features from each modality, reducing redundancy and improving efficiency. The weighted feature modality (Zhang et al., 2021a) assigns different weights to each feature based on its importance in emotion recognition. Additionally, attention mechanisms can prioritize salient features, using self-attention (Liu et al., 2022b) to refine intra-modal representations or cross-modal attention (Khan et al., 2024a) to capture interactions between modalities,

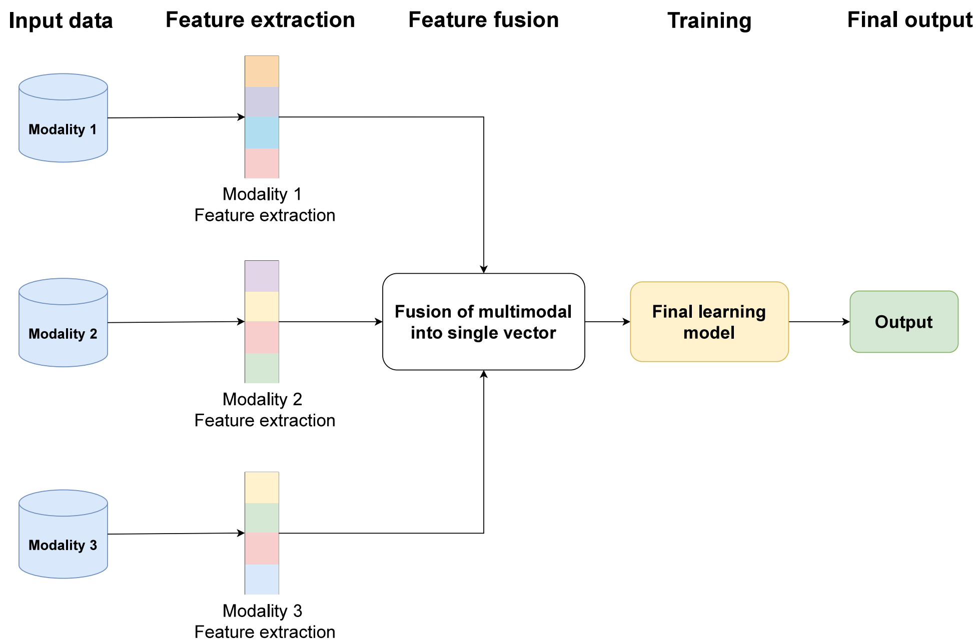


Fig. 7. General process of multimodal with feature fusion in SER.

enabling the model to focus on key emotional cues. This approach helps the model focus on the most relevant emotional cues.

This section explores various approaches to multimodal fusion in SER, categorized based on the stage at which data integration occurs. We examine four primary feature fusion methods: early fusion, deep fusion, late fusion, and hybrid fusion (D'Mello and Kory, 2012). Each method is analyzed regarding its model structure, advantages, and limitations. Furthermore, their effectiveness is evaluated based on WA and UA performance metrics reported in the literature, providing a quantitative comparison of different fusion techniques. WA and UA functions are formulated in Eqs. (1) and (2), considering class distribution to compute the weighted accuracy.

$$WA\ (\%) = \frac{\sum_{i=1}^{C} n_i a_i}{\sum_{i=1}^{C} n_i}, \tag{1}$$

$$UA\ (\%) = \frac{1}{C} \sum_{i=1}^{C} a_i, \tag{2}$$

where n_i is the number of samples in class i, a_i is the accuracy for class i, and C is the total number of classes.

WA reflects the overall accuracy by weighting each class based on its sample size, making it sensitive to class imbalances. Models optimized for WA tend to prioritize frequently occurring emotions, leading to higher accuracy for dominant classes but lower accuracy for underrepresented emotions (Geetha et al., 2024). In contrast, UA computes the mean accuracy across all classes, treating them equally regardless of sample distribution. High UA indicates balanced performance across all emotions, ensuring fair recognition of frequent and rare classes. However, optimizing for UA may reduce performance on majority classes, impacting overall prediction accuracy (Shah Fahad et al., 2021). By analyzing WA and UA together, a comprehensive assessment of fusion methods can be achieved, ensuring that the model is both effective in overall classification and fair across different emotional categories.

In addition to comparing the performance of feature fusion methods using accuracy metrics such as WA and UA, we also provide a comparative analysis of their computational complexity. Specifically, we evaluate the inherent architectural designs of early, late, deep, and hybrid fusion methods to estimate their computational cost. This dual assessment enables a more comprehensive understanding of the

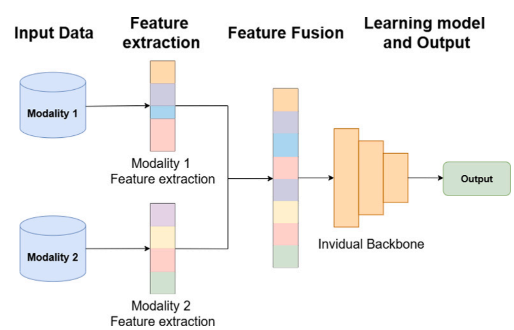


Fig. 8. Early fusion process illustration with two modalities utilizing simple concatenate operation.

trade-offs between predictive accuracy and resource efficiency, offering insights into the practical viability of each fusion approach.

Let f^{m1} and f^{m2} represent input features from two different modalities, which are fused using a fusion function $\mathcal{F}$. The fused features are then processed by the learning model $\mathcal{M}$, which outputs the result according to the objective model $\mathcal{C}$. We discuss the different fusion methods in the following subsections.

5.1. Early fusion

Early fusion occurs at the data level, typically in the input stage of each branch. In this approach, raw data samples from multiple modalities, such as acoustic features, linguistic content, or prosodic cues, are aligned and mapped into a common feature space using data alignment and transformation techniques. Fig. 8 illustrates an example pipeline of early fusion in SER involving two inputs: audio and dialogue. Early fusion in speech processing is a powerful technique that combines multiple data sources at the beginning of the processing pipeline, improving the model's ability to understand and interpret speech (Sun et al., 2018). While it quickly establishes relationships between modalities and effectively leverages emotional information from each, it also raises computational demands, mainly when dealing with

Table 4
Summary of early fusion method used for SER in different studies.

Paper	Feature types	Fusion method	Network architecture	Dataset	Performance	
					WA (%)	UA (%)
Shang and Fu (2024)	MFCCs and GloVe	Using multi-head self-attention mechanism and the improved sparrow search algorithm to combine feature	Obtain the ISSA-BiGRU-MHA method	IEMOCAP MELD MOSI CMU-MOSEI	76.52 71.84 66.72 62.12	77.13 69.12 64.33 61.74
Pentari et al. (2024)	eGeMAPS, Mel-spectrogram, ResNet, and graph-based	Using graph theory to extract statistical structural information from time series	Obtain the visibility graph random forest classifier in a leave-one-speaker-out cross validation scheme	EMO-DB AESDD DEMoS	77.80 70.00 79.1	– – –
Shahin et al. (2023)	MFCCs, ZCR, chroma, Mel-log spectrogram, and RMS	Using intelligent feature selection method based on a novel bio-inspired optimization algorithm	Grey wolf optimizer and K-nearest neighbor	Arabic Emirati accented speech RAVDESS SAVEE	83.16 56.59 69.01	83.25 56.64 72.34
Xie et al. (2023)	openSMILE, HOG, and WPD-based features	Combines acoustic features for frame-level properties, WPD-based features for multi-resolution decomposition, and temporal features	Random forest, grey wolf optimization based feature selection, and SVM	RAVDESS SAVEE EMOVO EMO-DB	86.97 88.79 89.24 95.29	86.75 88.58 90.05 96.12
Atila and Şengür (2021)	MFCCs, spectrogram, cochleagram, and fractal image features	Using attention-integrated 3D CNNs-LSTM, is developed for SER	Obtain attention-integrated 3D CNNs-LSTM, and cross-validation	RAVDESS RML SAVEE	96.18 93.32 87.50	– – –
Hu et al. (2021)	LSTM for text encoder, fully connected for audio	Multimodal features are concatenated and fed into GCN directly	Use a multimodal fused GCN to capture multimodal dependencies and speaker interactions	IEMOCAP MELD	64.46 57.94	– –

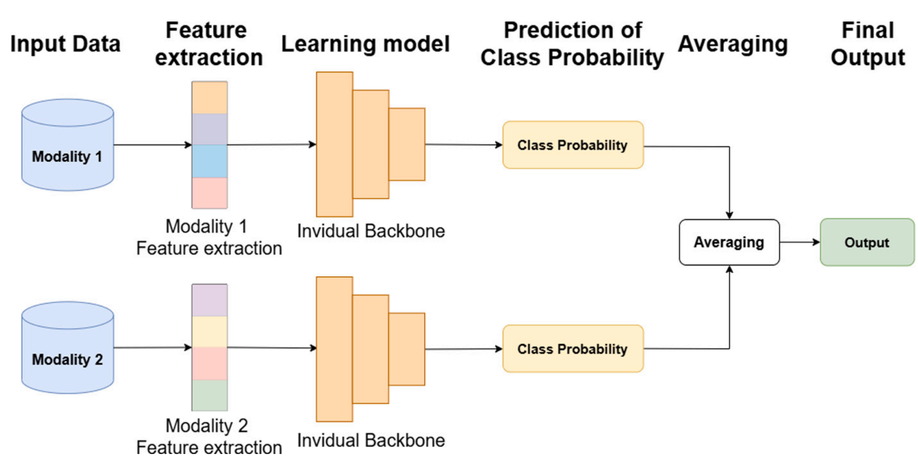


Fig. 9. Late fusion process illustration with two modalities utilizing class probability averaging.

complex multimodal data. Numerous investigations have incorporated deep fusion into early fusion (Vora et al., 2020). Eq. (3) represents how the fusion operation works in early fusion. The study on early fusion is described in Table 4.

$$output = C\left[\mathcal{M}\left(\mathcal{F}(f^{m1}, f^{m2})\right)\right]. \tag{3}$$

Early fusion tasks focus on processing input modalities to optimize information before feeding it into machine learning models or neural networks. In instance, Shang and Fu (2024), Shahin et al. (2023) used the early fusion for the LLDs feature from the audio modality. This helps capture fine-grained emotional cues at the early stage of the model pipeline, enhancing the model's sensitivity to subtle acoustic variations. On the other hand, Pentari et al. (2024) combined graph-based fusion with early fusion, allowing the system to extract structural and temporal dependencies across modalities. This integration enhances robustness and interpretability, especially in speaker-independent SER scenarios.

5.2. Late fusion

Late fusion in SER is a decision-level fusion approach during the prediction stage. Each modality (acoustic features, linguistic content, or physiological signals) has a separate decision-making processing branch in this method, as shown in Fig. 9. The final emotion classification is obtained by aggregating the predictions from each individual modality. The main goal of late fusion is to improve accuracy by leveraging the strengths of various views. Each view may capture different aspects of the data, and by fusing the results, the algorithm can achieve a more robust and accurate clustering outcome (Li et al., 2024a). However, it relies on unimodal information to predict results, which may lead to potential issues. Under certain conditions, important emotional cues might be overlooked or incorrectly detected due to limitations inherent in the individual modalities' data. The late fusion method is formulated in Eq. (4), where $\mathcal{M}_1$ and $\mathcal{M}_2$ correspond to the first and second learning models, respectively. Various studies on late fusion approaches in SER are summarized in Table 5, which provides a detailed comparison of methods, features, architectures, datasets, and performances used in

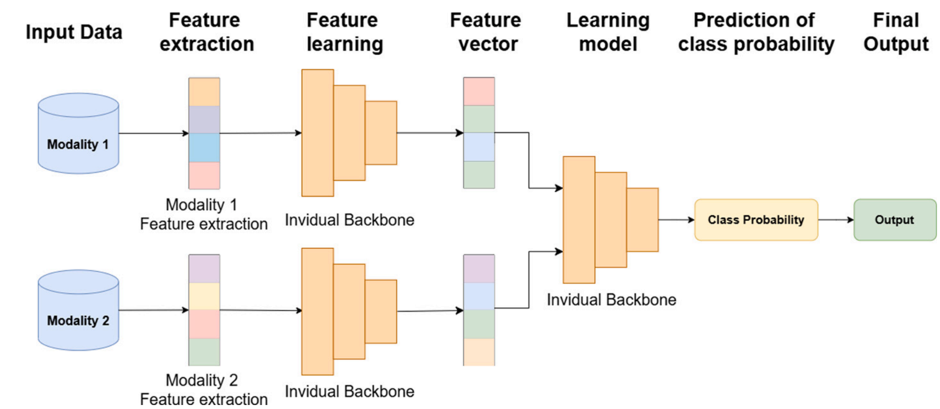


Fig. 10. Deep fusion process illustration with two modalities.

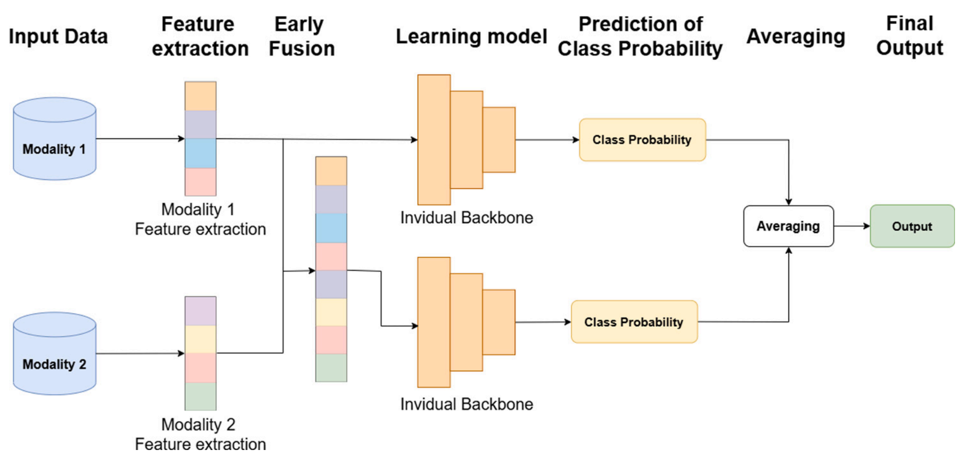


Fig. 11. Hybrid fusion process illustration with two modalities utilizing simple feature concatenation and class probability averaging.

recent research.

$$output = \mathcal{F}\left[\mathcal{C}\left(\mathcal{M}_1(f^{m1})\right), \mathcal{C}\left(\mathcal{M}_2(f^{m2})\right)\right]. \tag{4}$$

In late fusion, research focuses on designing architectures that integrate multimodal features after extraction to enhance SER. This approach enables models to exploit complementary spectral characteristics by capturing diverse frequency-band representations before decision-level fusion (Guo et al., 2023). Additionally, late fusion leverages the independent strengths of textual and acoustic modalities by aggregating their modality-specific predictions, thereby improving overall emotion detection performance (Khurana et al., 2024; Naderi and Nasersharif, 2023). For instance, Slimi et al. (2022) demonstrated that late fusion effectively combines the temporal modeling capabilities of LSTM and the spatial feature extraction strengths of CNN, leading to improved recognition accuracy.

5.3. Deep fusion

Deep fusion is implemented during the feature learning stage, where each modality is first processed independently through its backbone network to extract modality-specific feature vectors. As illustrated in Fig. 10, each modality undergoes separate feature learning through its backbone, and their extracted feature vectors are subsequently fused at an intermediate stage within the model. This fusion, often through concatenation, forms a unified representation that integrates complementary information from both modalities. The fused representation is then passed through a shared learning model to predict class probabilities and generate the final output. Deep fusion compensates for missing or weak features in one modality by leveraging the strengths of others, thereby enhancing robustness. Compared to early fusion, which combines features at the input level, deep fusion operates at a higher level, allowing for better alignment and reducing demands on input-level synchronization. However, deep fusion can suffer from high-dimensional feature spaces, increasing computational costs, and potential overfitting (Cairong et al., 2016). It may also introduce biases, such as overemphasizing dominant modalities or emotional categories due to corpus-specific traits. This can impact the model's generalization to other datasets or cultural contexts. Eq. (5) defines the formal process of deep fusion, where $\mathcal{M}_1$, $\mathcal{M}_2$, and $\mathcal{M}_3$ correspond to the respective learning models. Detailed methodologies and results are presented in Table 6.

$$output = \mathcal{C}\left[\mathcal{M}_3\left[\mathcal{F}\left(\mathcal{M}_1(f^{m1}), \mathcal{M}_2(f^{m2})\right)\right]\right]. \tag{5}$$

In deep fusion methods, diverse input data extraction techniques are leveraged to enhance SER by integrating multimodal information. Khan et al. (2024a) proposed a deep fusion network that combines spectrograms and text features using cross-attention and element-wise operations. Such deep fusion frameworks encompass a range of advanced feature combination strategies, including cross-attention mechanisms (Fu et al., 2024; Priyasad et al., 2023), multi-scale fusion (Yu et al., 2024), and graph-based integration (Meng et al., 2024), all of which have demonstrated improvements in SER performance. These approaches are designed to effectively capture both the global context and fine-grained emotional cues by aligning and integrating heterogeneous modalities.

Table 5
Summary of late fusion method used for SER in different studies.

Paper	Feature types	Fusion method	Network architecture	Dataset	Performance	
					WA (%)	UA (%)
Khurana et al. (2024)	Combines text features and audio features.	Fine-tuned RoBERTa-based text features and Inception-ResNet-V2 audio features	CNNs, BiLSTM, and cross attention and hybrid DNN	IEMOCAP	80.51	79.22
Guo et al. (2023)	Spectrogram, Mel-spectrogram, NGC	Feature fusion after frequency band division	ResNet34 with CBAM (attention mechanism)	ESD IEMOCAP	98.00 76.54	– –
Naderi and Nasersharif (2023)	Wav2Vec 2.0 transformer block, prosody features, MHA	To capture complex speech patterns and improve model performance	Uses a convolutional layer to reduce the dimensionality of Wav2Vec 2.0 features and a MHA layer to emphasize prosody and feature maps	IEMOCAP EMO-DB ShEMO Persian ESD TESS EMOVO SAVEE	74.16 – – – – – –	75.53 92.45 88.47 74.84 84.56 66.54 67.48
Li et al. (2023c)	Using DenseNet for visual, openSMILE toolkit for acoustic, and TextCNNs for text modalities	The model has three graphs, each containing information from only two modalities, which decreases the difficulty of multimodal fusion	Using ResGCN to handle GNNs' over-smoothing problem to improve the vanilla GATs	IEMOCAP MELD	67.90 61.30	68.07 58.37
Zhang et al. (2023a)	Prosody features, spectrogram features, and voice quality features	Using F-emotion algorithm to select features and build a parallel model for SER	Using the best features identified to produce distinct outcomes for each emotion category	EMO-DB RAVDESS	88.80 82.30	90.00 80.00
Slimi et al. (2022)	Output of LSTM for MFCCs, and CNNs for Mel-log spectrogram	Using decision-level fusion to combine different feature sets together	Using fusion of the output of two different models are CNNs, LSTM	RML	84.72	–
Liu et al. (2022a)	Feature extraction from variable utterances	Processes speech features through multi-feature fusion and combines them with a neural network classifier	Use graph pooling to combine node-level features from the GCNs into an utterance-level representation	IEMOCAP MSP-IMPRO	65.43 60.91	66.40 56.83
Zhang and Li (2022)	FBank features, MFCCs feature, and sound spectrogram feature	Processes speech features through multi-feature fusion and combines them with a neural network classifier	Hidden Markov models, and LSTM with Min, Average, Max pooling	100 courses on the MOOC platform	74.36	–
Xu et al. (2022)	Spectrogram, eGeMAPS, and MFCCs	Combines acoustic, temporal, and image speech features via a hybrid DL model	CNNs, LSTM, and hybrid DNN	EMO-DB IEMOCAP	91.25 72.02	90.61 73.42
Ibrahim et al. (2021)	The relation of the time steps in both forward and backward directions	Combined with the bidirectional approach and dimensionality reduction	Use ESN with bidirectional processing and SRP for emotion recognition	EMO-DB SAVEE RAVDESS FAU Aibo Emotion Corpus	89.45 80.28 89.11 63.30	91.08 81.10 88.89 32.23
Chen and Huang (2021)	The output of attention BiLSTM from log-Mel spectrogram, and MFCCs	Propose a dual-level model, called dual attention-based BiLSTM networks for SER	BiLSTM, dual attention, and dual ACRNN	IEMOCAP	–	70.29
Kwon et al. (2021)	Channel attention, and spatial attention	Combines attention weights based on SAM	CNNs, self-attention mechanism	EMO-DB IEMOCAP RAVDESS	94.00 77.00 81.00	93.00 78.01 80.00
Yao et al. (2020)	Obtain three individual models of LLDs-RNN, MS-CNNs, and HSFs-DNN	Fusion of more DL method can be carried out	LLDs-RNN, MS-CNNs, and HSFs-DNN	IEMOCAP	57.1	58.3

5.4. Hybrid fusion

In SER, hybrid fusion combines decision-level information from one modality with data-level or feature-level information from others, allowing for a more comprehensive integration of multiple modalities such as acoustic, linguistic, and visual features. Early fusion integrates raw features from different modalities, deep fusion captures intermediate representations at various network layers, and late fusion aggregates decisions from separate modalities (Zhang et al., 2021b). A hybrid fusion, whose function is expressed in Eq. (6), approach leverages the strengths of these methods to improve accuracy by integrating multimodal information, optimizing weight combinations across modalities, and capturing richer emotional cues (Wang et al., 2023a). In this formulation, $\mathcal{M}_1$ and $\mathcal{M}_2$ correspond to the respective learning models involved in the fusion process. However, this comes with increased model complexity and training challenges, necessitating careful tuning to balance performance and efficiency. As illustrated in Fig. 11, hybrid fusion is typically led by at least one primary branch, while the other modal branches contribute supplementary information to support the final task. Table 7 provides a detailed overview of hybrid fusion

Table 6
Summary of deep fusion method used for SER in different studies.

Paper	Feature types	Fusion method	Network architecture	Dataset	Performance	
					WA (%)	UA (%)
Khan et al. (2024a)	Spectrol feature and text features	Deep fusion network a cross-attention mechanism and element-wise operations	One-dimensional CNNs and GRUs network cross attention	IEMOCAP MELD	77.20 –	76.56 72.30
Fan et al. (2024)	Pairwise fusion of video, audio, and text features using a graph-based model	Proposed a novel pairwise modalities fusion for SER	GCN-based dual-modal fusion with BiLSTM encoding	IEMOCAP MELD	69.57 66.05	69.34 67.13
Yu et al. (2024)	Output of three backbone of CNNs with difference pooling	Multi-scale feature fusion	Multi-dimensional feature extraction with CNNs	IEMOCAP CREMA-D RAVDESS	77.50 64.62 85.51	77.63 64.73 85.76
Meng et al. (2024)	Features are extracted using RoBERTa (text), openSMILE (audio), and 3D-CNNs (video)	Multi-message passing with a heterogeneous graph, dynamically assigning weights based on node homogeneity	Multivariate message passing GCNs with BiLSTM for context modeling, and self-attention for feature fusion	IEMOCAP MELD	68.90 60.70	68.00 59.30
Fu et al. (2024)	Sequentially encoding text, audio, and video messages and inputting them into a Bi-GRU to maintain context and order	MHA mechanism and GNNs to fuse multimodal data and capture cross-modal context information	MM DialogueGAT, consisting of sequential context encoding (using Bi-GRU), cross-modal integration, interlocutor graph encoding with GATs, and a sentiment classifier	IEMOCAP	81.46	81.99
Zhang et al. (2023b)	MFCC and log-Mel spectrogram	Concatenation after CBAM attention	ResGCN, AlexNet, and CBAM attention	CASIA IEMOCAP RAVDESS	92.20 70.10 79.80	92.40 69.90 69.90
Priyasad et al. (2023)	Audio features from a fine-tuned Wav2Vec model and text features from BERT	The model integrates information over time using memory modules	Wav2Vec 2.0 and BERT embeddings use attention-enhanced memory blocks with cross-attention to capture complementary audio-text information	IEMOCAP	76.80	77.30
Yuan et al. (2023)	Textual features are extracted using TextCNNs, acoustic features using the openSMILE, and visual features using DenseNet	The proposed method, relational bilevel aggregation GCNs, uses a multimodal fusion approach	Using the graph generation module, the similarity-based cluster building module, and the bilevel aggregation module	IEMOCAP MELD	71.43 62.67	– –
Chen et al. (2023)	Frame-level deep spatial and temporal features; Utterance-level deep features spectrogram	Multi-scale deep emotional feature fusion using MHA	CNNs, BiLSTM, and MHA mechanism	IEMOCAP RAVDESS	81.60 88.88	79.32 87.85
Li et al. (2023a)	Two cross-modal fusion attention blocks	Using cross-modal attention to extract emotion-related features from audio and text	The model uses Wav2Vec 2.0, RoBERTa-base for sound, text encoder, and CMFA	IEMOCAP	81.20	80.50
Abdusalomov et al. (2023)	MFCC feature encoder, and BiLSTM encoder	Employed a dual-strategy approach for feature extraction	Multi-dimensional feature extraction with CNNs	CMU-MOSEI RAVDESS	– –	93.20 95.30
Wen et al. (2022)	TFCNNs features, and VGGish features	Using self-labeling with feature transfer for SER from spectrogram, and log Mel-spectrum	TFCNNs, CNNs-BiGRU, VGGish, and self-labeled	IEMOCAP EMO-DB SAVEE	73.03 83.14 52.09	– – –
Xu et al. (2021)	MFCCs with two parallel convolutional layers to extract textures from the horizontal and vertical respectively	Proposed head fusion based on the MHA mechanism	Implemented an attention-based CNNs	IEMOCAP	76.18	76.36

approaches in recent studies, comparing the techniques, datasets, and resulting accuracies to highlight the effectiveness and variations in hybrid fusion method across different research.

$$output = C\left[\mathcal{F}\left(\mathcal{M}_1\left(f^{m1}\right), \mathcal{M}_2\left(\mathcal{F}(f^{m1}, f^{m2})\right)\right)\right]. \quad (6)$$

Hybrid fusion offers distinct advantages in multimodal learning by uniting the strengths of early and late fusion methods. This approach minimizes information loss and captures more comprehensive feature representations across modalities. By enabling simultaneous low-level feature integration and high-level decision alignment, hybrid fusion enhances contextual understanding and semantic depth. It excels at modeling nonlinear relationships between modalities through cross-modal attention and graph-based reasoning, significantly boosting classification performance (Liu et al., 2024b; Hong et al., 2024; Thanh et al., 2024). Moreover, the hybrid fusion method's adaptability to diverse data types and resilience to noise or missing modalities make hybrid fusion a robust and versatile solution for SER and other multimodal tasks.

5.5. Comparisons of feature fusion methods

5.5.1. Accuracy comparison

A direct and comprehensive experimental comparison between of fusion methods is presented by Khan et al. (2024a). The authors systematically evaluate early fusion, late fusion, and their proposed deep fusion method as final integration methods. The results on the IEMOCAP dataset demonstrate the superior accuracy of deep fusion, demonstrating its effectiveness in capturing rich cross-modal dependencies for emotion recognition.

Their results provide clear insights into the relative accuracy of these methods, as visualized in Fig. 12. Early fusion results in the lowest accuracy, with a WA of 75.50% and a UA of 74.80%. This involved concatenating initial feature representations before fitting them into a joint model. Late fusion performed second best, yielding a WA of 76.10% and a UA of 74.84%. This typically involves combining the outputs or decisions from separately trained modality-specific backbones. Since the proposed deep fusion approach involved combining intermediate layer representations with auxiliary losses, it achieved the highest accuracy with WA of 77.20% and UA of 76.56%.

5.5.2. Computational cost comparison

In parallel with the accuracy comparison of feature fusion methods, Table 8 evaluates the computational cost of representative models for each fusion method: early fusion (Hazarika et al., 2018), late fusion (Ai et al., 2025), deep fusion (Hazarika et al., 2020; Lin et al., 2023; Peng et al., 2023; Li et al., 2024b), and hybrid fusion (Zadeh et al., 2017; Tsai et al., 2019). The results show that early fusion has the lowest computational cost, with minimal parameter size and floating point operations per second (FLOPs). Late fusion incurs slightly higher resource usage but remains efficient overall. Deep fusion methods result in significantly larger models, with parameters ranging from 11M to 24M and FLOPs between 319M and 545M. Hybrid fusion methods are the most computationally expensive, with parameter sizes exceeding 150M and FLOPs ranging from 2.31G to 3.40G. These findings demonstrate an apparent increase in computational demand from early to hybrid fusion, emphasizing the trade-off between fusion complexity and resource efficiency.

We observe differences in computational complexity among feature fusion methods, as indicated by their architectural design, measured through FLOPs and model parameter count. Early fusion methods are generally lightweight because they merge modality features at the input level and process them jointly through a unimodal approach, avoiding the need for complex interaction modeling. Late fusion also remains efficient by processing each modality independently and combining the outputs at the decision level using simple techniques like averaging or voting, resulting in moderate computational demands. In contrast, deep and hybrid fusion approaches integrate multimodal information at intermediate or final stages of the network, often relying on mechanisms such as cross-modal attention, multi-stage representation learning, or tensor-based fusion. These techniques introduce substantial computational overhead, increasing FLOPs, inference time, and memory consumption. The architectural depth and interaction complexity inherent in deep and hybrid fusion models largely explain their significantly higher computational cost than early and late fusion strategies.

While deep and hybrid fusion approaches often yield better accuracy due to their ability to capture rich multimodal interactions, they also introduce substantial computational overhead. These approaches frequently utilize cross-modal attention, multi-stage encoding, or tensor-based interactions to model intricate relationships across modalities. While these methods enhance accuracy by capturing richer representations, they significantly increase model parameters, memory requirements, and inference time. This underscores a consistent trade-off between improved performance and computational complexity. Consequently, selecting a fusion method requires careful consideration of both performance objectives and the constraints of the deployment environment, especially in real-time or resource-constrained scenarios.

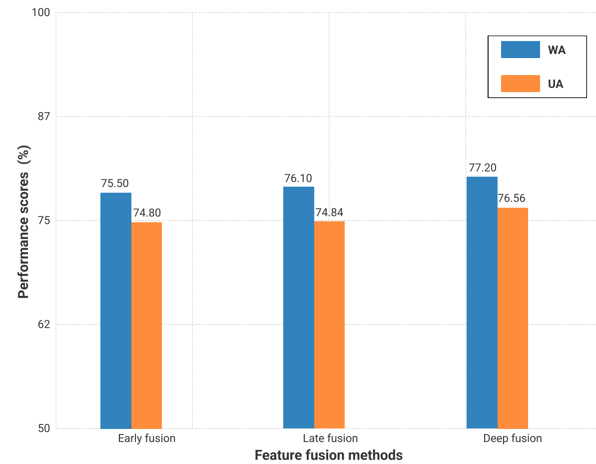


Fig. 12. Accuracy comparison of early, late, and deep feature fusion methods on the IEMOCAP dataset. This figure is generated based on the experimental results reported by Khan et al. (2024a).

5.6. The evaluation framework for assessing multimodal fusion methods

We adopt a structured evaluation framework, which is visualized in Fig. 13, to systematically evaluate and compare the different multimodal feature fusion methods discussed in this review: early, late, deep, and hybrid fusion. Establishing such frameworks is crucial for rigorous comparison in multimodal analysis, providing a consistent methodology to assess performance across different techniques and datasets. Building upon foundational research like Peña et al. (2023) establishes a base to perform a comparative evaluation of fusion methods to demonstrate how they adapt to individual modalities' quality differences and evaluate their performance. Our framework is enhanced to incorporate advancements such as integrating transformer-based models for feature representation.

5.6.1. Data collection and preparation

The data collection and preparation stage forms the foundation of the multimodal SER framework. It begins with acquiring a comprehensive emotion dataset that includes three core modalities: original audio, transcribed text, and visual frames extracted from videos. After being collected, each modality undergoes a modality preparation process to ensure quality and consistency. This step ensures that all inputs are standardized and ready for feature extraction. After cleaning, the dataset is strategically divided into two parts: a training set for model development and hyperparameter optimization, and an independent testing set reserved exclusively for final evaluation. This separation is crucial to ensure fair assessment and prevent data leakage, particularly in real-world deployment scenarios.

Table 7
Summary of hybrid fusion method used for SER in different studies.

Paper	Feature types	Fusion method	Network architecture	Dataset	Performance	
					WA (%)	UA (%)
Song et al. (2024)	MFCCs, spectrogram, and raw waveform	Using multi co-attention mechanism to enhance the acoustic characteristics	CNNs, BiGRUs, and self-supervised Wav2Vec 2.0 based models	IEMOCAP	–	69.08
Yan et al. (2024a)	Temporal features Spatial features	Combines temporal and spatial features through an adaptive graph structure and BiLSTM	BiLSTM, LGCN, and adaptive adjacency matrix	MSP-IMPROV IEMOCAP	58.35 66.82	56.47 64.21
Liu et al. (2024b)	Logarithmic Mel-spectrograms: deltas and delta-deltas of log-mel spectrograms	Multi-scale self-attention with cross-scale attention fusion	Multi-view attention network integrated with a multi-time view BiLSTM and a diffusion joint loss mechanism	EMO-DB IEMOCAP CASIA	86.87 70.74 93.65	86.60 70.74 91.13
Hong et al. (2024)	Text features are extracted using BERT; visual and audio features are processed with LSTM	Using cross-modal dynamic transfer learning that filters and aligns features from each modality based on their confidence scores	The architecture filters low-confidence modalities, and then applies bidirectional knowledge transfer to fine-tune the fusion for each modality pair	CMU-MOSEI IEMOCAP	47.25 61.30	55.66 61.47
Thanh et al. (2024)	Wav2Vec 2.0 and pitch encoder features	Use a pitch encoder and cross-attention to align pitch features with Wav2Vec 2.0 acoustic features	The architecture fuses pitch and acoustic features using cross and self-attention	ViSEC ASVP-ESD Thai SER	71.90 84.77 87.73	72.72 85.40 87.47
Li et al. (2023b)	ZCR, log energy, short-time energy, and MFCCs	Integrates a CNNs-based multiperspective aware module with a frame-level fine-grained fusion method	Combine two types of acoustic features: frame-level features and segment-level features	IEMOCAP	72.19	72.88
Santoso et al. (2022)	Acoutics feature, text features	Using confidence measure to mitigate the SER performance deterioration due to SER errors	BiLSTM, LSTM, and self-attention	IEMOCAP	76.60	76.80
Shixin et al. (2022)	Acoustic temporal features are fused with text emotion features	The method fuses acoustic features (via TGFE with CNNs-BiLSTM) and text emotion features (via BERT and BiLSTM)	Using transfer learning with RoBERTa and ResNet-V2	CMU-MOSEI IEMOCAP MELD	99.20 72.80 63.80	– – –
Tang et al. (2022)	Audio features are processed using BiLSTM and MSA, text features are processed with GRUs	Fuse and utilize information within acoustic and textual modalities	Using CNNs, BiLSTM, audio–text-interactional-attention with ArcFace loss	IEMOCAP	82.40	80.60
Zou et al. (2022)	MFCCs, spectrograms, and Wav2Vec 2.0 features	Utilized a co-attention mechanism to integrate the various acoustic features	Using three encoders for acoustic features: CNNs for spectrogram, BiLSTM for MFCCs, and Wav2Vec 2.0 for raw audio	IEMOCAP	71.64	72.70
Lian et al. (2021)	Combining text, audio, speaker, and cross-modal features from different modalities through transformers	Propose to fuse multimodal features via the attention mechanism	Conversational transformer network architecture combines single-modal and cross-modal transformers, speaker-aware audio-text-speaker fusion, and MHA-GRU	IEMOCAP MELD	83.60 62.00	83.30 60.50
Ho et al. (2020)	The first level generates a representation of audio and text RNN features, while the second level uses MHA to fuse them	Leveraging temporal-contextual features and fusing two modalities with different lengths	Multi-level multi-head fusion attention-based RNN with global average pooling	CMU-MOSEI IEMOCAP MELD	99.19 74.33 59.94	99.19 73.23 63.26
N. and Patil (2020)	Raw audio waveform, and word embeddings	Integrate audio and text encoder, enabling mutual attention for improved emotion recognition	Using cross-modal attention with 1D-CNNs and BiLSTM	IEMOCAP	–	72.82

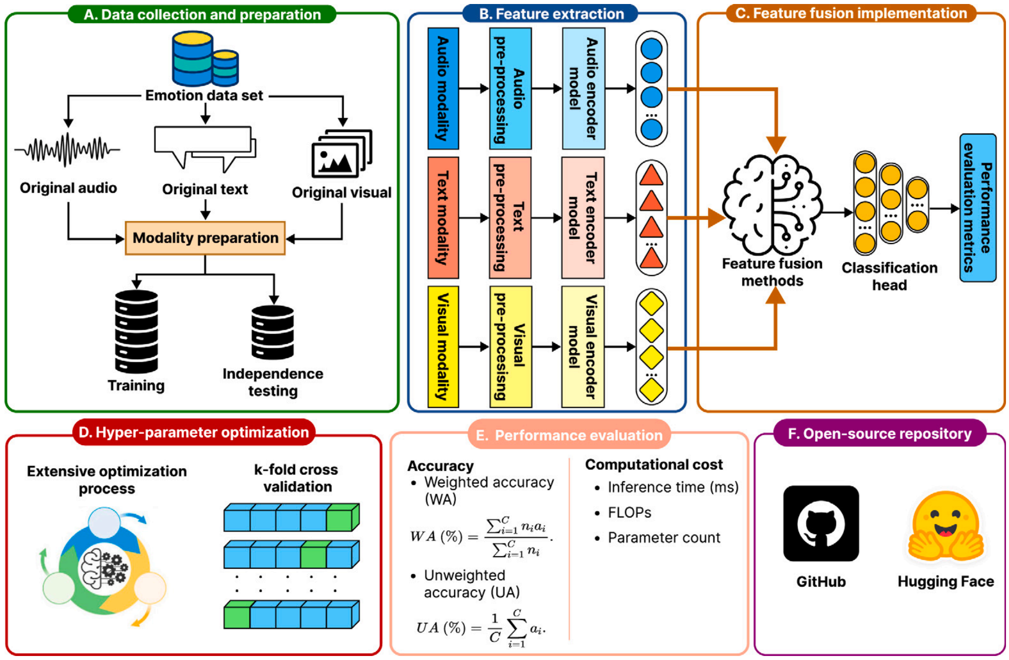


Fig. 13. The overview of evaluation framework of feature fusion method in multimodal SER.

Table 8
Computational cost comparison of early, late, deep, and hybrid fusion methods on the MELD dataset. The symbol (↓) shows that the lower cost is better.

Paper	Feature fusion	Params (↓)	FLOPs (↓)
Hazarika et al. (2018)	Early fusion	1.38M	1.32M
Ai et al. (2025)	Late fusion	2.10M	15.27M
Hazarika et al. (2020)	Deep fusion	11.30M	544.79M
Lin et al. (2023)	Deep fusion	20.68M	428.75M
Peng et al. (2023)	Deep fusion	24.13M	458.37M
Li et al. (2024b)	Deep fusion	16.82M	319.08M
Zadeh et al. (2017)	Hybrid fusion	179.20M	3.40G
Tsai et al. (2019)	Hybrid fusion	159.52M	2.31G

5.6.2. Feature extraction

The feature extraction stage is a critical component in the multimodal learning pipeline, focusing on transforming raw data from each modality into meaningful high-level representations. This stage is organized into three parallel streams based on the audio, text, and visual modalities. Each modality first undergoes its pre-processing phase, tailored to the nature of the input. For the audio modality, pre-processing may include several steps, such as silence removal, normalization, and resampling, to enhance signal quality and prepare it for analysis. In the text modality, tokenization and text normalization are applied to clean and structure the input, making it suitable for embedding-based models. The visual modality involves frame sampling and face cropping, which help isolate the most informative regions from video data.

After pre-processing, each modality is passed through a dedicated encoder model designed to extract high-level features. The audio encoder, such as Wav2Vec 2.0 or CNN-based models, captures acoustic features like pitch, tone, and spectral patterns. The text encoder, for instance, Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al., 2019) or RoBERTa (Liu et al., 2019), learns syntactic and semantic representations from the linguistic input. Meanwhile, the visual encoder, such as Residual Network (ResNet) (Targ et al., 2016) or VGG (Sengupta et al., 2019), focuses on extracting spatial features from facial expressions or visual scenes. These high-dimensional feature vectors from each modality are then forwarded to the fusion stage for multimodal integration.

5.6.3. Feature fusion and implementation

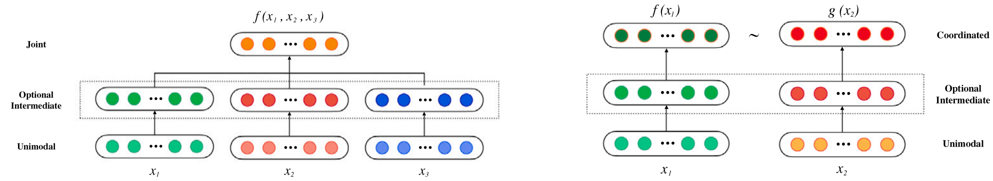
After extracting features using dedicated encoders, the next step is feature fusion and implementation, which aims to integrate information from audio, text, and visual representations into a unified embedding for emotion recognition. The feature vector, which captures acoustic cues, linguistic content, and visual expression, is passed into a fusion module that learns inter-modality relationships and complementary patterns. Feature fusion methods include early, late, deep, and hybrid fusions. This framework passes the fused representation into a classification head for final emotion prediction. The model is trained and optimized using performance metrics such as accuracy and inference time. This ensures that the fusion method effectively captures complementary information across modalities for robust multimodal SER.

5.6.4. Hyper-parameter optimization

This stage involves extensive optimization to fine-tune the model's performance by selecting the most effective combination of hyper-parameters. Parameters such as learning rate, batch size, dropout rate, number of MLP layers, and feature dimensions are systematically adjusted. To ensure robust generalization, the framework employs k-fold cross-validation, where the training set is split into k subsets, and the model is trained and validated k times. This process helps mitigate overfitting and provides a more reliable estimate of the model's performance.

5.6.5. Performance evaluation

In the performance evaluation stage, the framework rigorously assesses the effectiveness and efficiency of the multimodal SER model. Accuracy is evaluated using two key metrics: WA, which considers the distribution of classes to account for any imbalance, and UA, which treats all classes equally to provide a balanced performance measure across emotion categories. Beyond accuracy, computational



(a) Joint representation. (b) Coordinated representation.

Fig. 14. Structure of joint and coordinated representations.

Table 9
A summary of the advantages and disadvantages of data representation.

Data representation	Paper	Advantages	Disadvantages
Joint representation	Slimi et al. (2022), Liu et al. (2022a), and Xu et al. (2022)	Integrates audio and text features to capture cross-modal dependencies, enhance performance, and simplify processes through end-to-end learning	High computational cost, risks overfitting with limited data, requires complex architecture design and tuning
Coordinated representation	Guo et al. (2023), Naderi and Nasersharif (2023), Zhang et al. (2023a), Zhang and Li (2022), Khurana et al. (2024), Ibrahim et al. (2021), Chen and Huang (2021), Kwon et al. (2021), and Yao et al. (2020)	Enables different modalities to interact and complement each other throughout processing allows better feature interaction	Limited to capture cross-modal relationships, heavy dependence on effective fusion method, and potential latency issues in real-time applications

complexity is also measured to ensure the model's practicality in real-world scenarios. The computation complexity includes inference times, FLOPs, and the parameter count. These metrics comprehensively show the model's predictive performance.

5.6.6. Open-source repository

The complete framework, encompassing data pre-processing scripts, modality encoders, fusion techniques, training pipelines, and evaluation tools, is designed to be openly accessible, promoting reproducibility and fostering further research. After running the framework, users can upload it to platforms such as GitHub[1] or Hugging Face.[2] This open-source approach aims to empower the community to develop and benchmark multimodal SER systems effectively.

6. Review of fusion technologies for SER

6.1. Data representation

Data representation is a critical component of ML, particularly in SER. It transforms raw speech data into meaningful features for ML models to analyze and classify emotions. While traditional methods rely primarily on handcrafted features with limited generalization capabilities, DL has revolutionized this process. Indeed, advanced DL models can automatically learn representations from raw audio signals, leveraging acoustic features and para-linguistic cues to capture the intricate nuances of emotional expression in speech signals (Latif et al., 2023). Furthermore, an effective data representation method should capture essential information and create an implicit vector for multimodal data. High-quality representation learning retains critical details, enhancing performance in downstream tasks. It must also address noise between modalities, data loss, real-time processing, and efficiency challenges (Bengio et al., 2013).

SER systems employ two primary types of data representation: joint and coordinated (Baltrušaitis et al., 2019). Joint representation combines multiple features into a unified space for analysis, while coordinated representation processes different features separately but aligns them to enhance emotional nuance recognition. SER systems can improve their accuracy and robustness by utilizing these deep representation techniques, leading to more effective real-time emotion detection across various applications. The joint and coordinated representation structure is visualized in Fig. 14. Table 9 is the distribution of the related paper, the advantages and disadvantages of joint and coordinated representation.

6.1.1. Joint representation

In joint representation, the mathematical formulation is expressed in Eq. (7).

$$x_m = f(x_1, x_2, \dots, x_n), \tag{7}$$

where x_m represents the multimodal representation, $f(x)$ is a function that combines unimodal representations (such as a deep neural network, restricted Boltzmann machine, or a recurrent neural network), x_1, x_2, …, x_n are the individual unimodal representations.

Joint representation combines multiple features derived from speech signals, including spectral, prosodic, and linguistic information, and passes them through dedicated encoders. It is possible to extract multi-level acoustic features, such as Mel-frequency cepstral coefficients (MFCCs) and spectrograms, which are vital for capturing the emotional subtleties in speech signals. These features provide complementary information at different levels, helping models better understand emotional content. By employing co-attention mechanisms, ML models can effectively fuse these diverse acoustic features, capturing intricate relationships between various aspects of speech signals (Zou

[1] https://github.com/
[2] https://huggingface.co/

et al., 2022). Additionally, emotion-pair-based frameworks demonstrate that considering the proximity of emotions within a dimensional space can improve the precision of bi-classification results (Chen et al., 2018b). Joint learning that addresses emotion classification and cause detection can exploit shared feature representations, leading to enhanced overall result (Ma et al., 2017).

6.1.2. Coordinated representations

Coordinated representation can be mathematically expressed in Eq. (8). In this formulation, each modality, represented by x_1 and x_2, has its projection function, denoted by $f(x)$ and $g(x)$. They map the unimodal inputs into a shared multimodal space. The coordination between these projected multimodal representations, indicated by symbol ($\sim$), ensures that the representations are aligned or coordinated within the shared space. This coordination can be achieved through various techniques, such as minimizing the cosine distance between projections, maximizing their correlation (Morcos et al., 2018), or enforcing a partial order between them (Desai et al., 2023).

$$f(x_1) \sim g(x_2). \tag{8}$$

Coordinated representations operate by projecting inputs from various modalities, such as acoustic and textual features, into a shared multimodal space where the representations are aligned and can enhance one another. A dedicated encoder processes each modality, transforming the inputs into high-level features. These features are aligned within the shared space through coordination mechanisms. Techniques such as cosine similarity minimization, correlation maximization, and partial order enforcement are commonly employed for this purpose (He et al., 2016; Chen et al., 2023a). Once the features are aligned, they are fused, typically through concatenation or co-attention mechanisms, to create a unified feature set that integrates complementary information.

6.2. Data translation

Data translation is fundamental and involves interconverting between different data modalities, aiming to compensate for information lost in one modality by incorporating insights from another. This process helps capture complementary information across various modalities. Technologies commonly used for multimodal data translation include neural networks, graphical models, and generative adversarial networks (GANs). Neural networks are widely adopted due to their strong learning capabilities, allowing multimodal data to be mapped into a shared semantic space. Graphical models, on the other hand, represent different modalities as graph structures and leverage their propagation and aggregation abilities to translate data between modalities. GANs, an emerging methodology, use an adversarial process between a generator and a discriminator to handle multimodal data translation. The generator maps data from different modalities into the semantic space, while the discriminator evaluates the accuracy of this translation. Through iterative optimization of the generator-discriminator interaction, GANs achieve improved performances in translating multimodal data. These approaches enable the seamless integration of diverse data types, leading to more accurate and comprehensive models.

Multimodal data translation methods encompass various approaches that, despite their modality-specific implementations, share fundamental principles in handling cross-modal conversion. These methods can be categorized into two main approaches: example-based models that utilize predefined mappings and dictionary-based translations between modalities and generation models that enable dynamic, real-time translation of multimodal information. While example-based models excel at direct translations using established patterns, generation models provide seamless, dynamic translations. This framework of approaches provides a comprehensive foundation for addressing various multimodal translation challenges (Baltrušaitis et al., 2019). The illustrator structure of example-based and generative approaches in multimodal translation is visualized in Fig. 15.

6.2.1. Example-based approaches

Example-based approaches are limited by their training data, often called the dictionary. These approaches can be classified into two types: retrieval-based and combination-based. Retrieval-based models use the retrieved translations directly without making any modifications, while combination-based models employ more complex rules to generate translations by combining multiple retrieved examples.

Retrieval-based models offer a straightforward approach by leveraging a database of pre-labeled emotional speech samples (Asai et al., 2023). These models identify the most semantically similar example within this emotional dictionary and assign its corresponding emotion label to the input speech sample. This retrieval process can occur in two primary ways: using only acoustic features extracted from the speech signal (unimodal retrieval) or incorporating additional contextual information or data from other modalities, such as text or facial expressions (multimodal/contextual retrieval). This latter approach enriches the semantic representation of the input, potentially leading to more accurate emotion recognition. Essentially, retrieval-based models for SER exploit the power of similarity search within a structured emotional database, with the option to enhance accuracy through the fusion of multimodal data.

Building upon the foundation of retrieval-based models, combination-based approaches enhance SER by retrieving relevant examples from the emotional dictionary and combining them strategically. The strategy stems from the observation that emotional expressions in speech often exhibit recurring patterns and structures. Combination-based models leverage these patterns by employing predefined rules, often handcrafted or based on heuristics, to fuse multiple retrieved examples into a more accurate representation of the target emotion. This fusion process allows for a richer interpretation of emotional states, moving beyond simple matching to capture the complexities inherent in human vocal expression.

6.2.2. Generative approaches

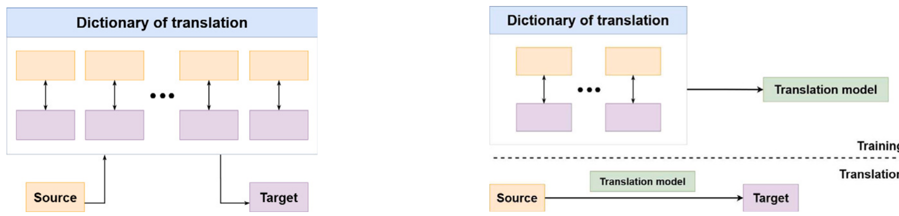
Generative approaches in multimodal translation enable the underlying models to perform translations across different modalities, even when given only a unimodal input. This task is complex because it requires the model to understand the source modality and generate a coherent sequence or signal in the target modality. Evaluating such models is also challenging due to the wide range of correct outputs.

Grammar-based models use a predefined grammar to generate a target modality. They start by detecting high-level concepts within the source modality, organized and incorporated into a structured generation process based on the predefined grammar. This approach allows for controlled and coherent outputs, which is particularly useful in settings where structured, rule-based output is required.

Encoder–decoder models process audio input, converting it into a sequence of features that can capture emotional content. End-to-end trained encoder–decoder models are among the most widely used techniques for multimodal translation. This setup typically involves encoding the temporal and spectral information from speech and decoding it into categorical or continuous emotion labels. The encoder–decoder model is primarily used for text generation but can generate images and sounds. The decoder is typically implemented with an RNNs or LSTM network, using the encoded representation as the initial hidden state.

Continuous generation models use advanced neural network architectures to create dynamic representations of language and emotion, enabling more nuanced outputs. They are particularly effective for sequence-to-sequence tasks in SER, where they process audio input in real-time and produce outputs that classify or represent emotional states at each time step. This approach is precious for applications such as tracking emotions in spoken dialogue and analyzing emotional states in audio streams in real-time.

Wen et al. (2022) enhanced the self-labeling method's effectiveness by combining a time-frequency CNN (TFCNN) with features from a pre-trained VGGish model on large audio datasets, addressing limitations



(a) Example-based approach. (b) Generative approach.

Fig. 15. Structure of multimodal data translation.

from small SER datasets through transfer learning. Hong et al. (2024) proposed an example-based approach, where text, audio, and visual data features are transformed into unified embeddings and aligned through cross-modal knowledge transfer, enhancing consistency across modalities. Li et al. (2023b) enhanced SER by using a multiperspective awareness module to capture emotional nuances at local, frame, and global levels and a fine-grained fusion method that used frame-level attention to integrate diverse emotional features, improving classification accuracy.

6.3. Attention mechanism

In multimodal SER, attention mechanisms are precious for integrating and prioritizing information from multiple data sources. Each modality offers unique insights: audio data captures vocal tones, energy, and pitch changes, while text data (e.g., transcriptions) provides semantic content that can reveal subtle emotional cues. In some systems, visual data, such as facial expressions, can further enrich the emotional context. In this review, we explore various attention-based approaches that enhance the performance of SER models by effectively merging these diverse modalities.

6.3.1. MHA mechanism

MHA, originating from the transformer model, is highly effective for SER as it enables the model to capture different emotional cues within each modality. By assigning multiple attention heads to focus on different parts of the input, the model can capture diverse emotional expressions, such as shifts in tone or semantic meaning. This parallelized focus improves the presentation of nuanced emotions, as it can consider multiple emotional indicators simultaneously, enhancing both intra-modality and inter-modality interactions. MHA operates using three key components: the Query (Q), Key (K), and Value (V) matrices. Q represents the vector that asks a "question" about the input data, determining how much focus each input token should give to other tokens. K serves as the reference vector against which queries are matched, helping to compute the similarity or relevance between tokens. Finally, V contains the actual information being passed through the attention mechanism, where the attention scores derived from Q and K are applied to V to determine its contribution to the output. MHA is formed by the interaction of multiple attentions, which can pay attention to different positional information. The formula can be written as Eqs. (9) and (10).

$$MultiHead(Q, K, V) = Concat(h_1, \dots, h_h)W^O, \tag{9}$$

where multiple attention heads (h_1 to h_h) are concatenated and then multiplied by the output weight matrix W^O to produce the final output.

$$h_i = Attention(Q \cdot W_i^Q, K \cdot W_i^K, V \cdot W_i^V), \tag{10}$$

where h_i represents a single attention head, the queries Q, keys K, and values V are multiplied by their respective weight matrices W_i^Q, W_i^K, W_i^V before applying the attention mechanism.

Shang and Fu (2024) employed the MHA mechanism in their proposed ISSA-BiGRU-MHA method for emotion recognition. As described by Naderi and Nasersharif (2023), this technology emphasized the most critical aspects of prosody and convolution feature maps by using global prosody as the query input to prioritize relevant features for enhanced analysis dynamically. To capitalize on the strengths of diverse feature types, Chen et al. (2023) and Tran et al. (2023a,b) introduced a MHA mechanism integrating deep representations from feature levels. Xu et al. (2021) demonstrated that MHA enables diverse speech pattern capture across attention heads, where head fusion enhanced robustness and generalization to noise, notably under conditions like Discrete-Short noise. Lian et al. (2021) proposed using a BiGRU layer to capture contextual information in both directions, combined with a MHA mechanism to emphasize critical contextual utterances. Leveraging the temporal-contextual features of each modality, Ho et al. (2020) effectively fused two modalities with varying temporal lengths using a proposed two-level attention mechanism.

6.3.2. Cross-modality attention

Cross-modality attention helps models selectively emphasize sensory information across multiple modalities. It extends the traditional attention mechanism by allowing Q to come from one modality and K, V from another modality (You et al., 2020). This setup facilitates learning how one modality influences or complements another. By integrating complementary information from each modality, cross-modality attention enables SER systems to interpret emotions more contextually and accurately. Eq. (11) describes the cross-modality attention mechanism when integrating text and audio features.

$$\text{Attention}_{\text{cross}}(Q_1, K_2, V_2) = \text{softmax}\left(\frac{Q_1 \cdot K_2^T}{\sqrt{d_k}}\right)V_2. \tag{11}$$

In Eq. (11), Q_1 is sourced from the text modality, representing the semantic context of words or sentences by a text encoder like RoBERTa or BERT. K_2 and V_2 are derived from the audio modality, capturing features like prosody, pitch, or spectral information through models like Inception-ResNet or openSMILE. The operation computes the attention weights by taking the scaled dot product of Q_1 and K_2, normalizing the results with a softmax function, and then applying these weights to V_2.

Khurana et al. (2024) leveraged cross-attention to align and enhance emotional context across modalities. Khan et al. (2024a) applied cross-modality attention to align audio intonations with corresponding emotionally charged words, allowing the model to capture richer emotional representations. Fu et al. (2024) introduced a cross-modal MHA mechanism to integrate information across different modalities in their emotion recognition model. Each modality (text, audio, or visual) in this setup has distinct weights calculated by separate attention heads. Li et al. (2023a) also employed cross-modal attention to capture emotion-specific features from audio and text, effectively aligning emotional nuances across modalities. Thanh et al. (2024) and Le et al. (2024) leveraged cross-modal attention to capture emotion-specific features from audio and text, aligning emotional nuances effectively across modalities. N. and Patil (2020) implemented cross-modal attention, allowing features from the audio encoder to attend to those from the text encoder and vice versa.

6.3.3. Dual attention networks

The dual attention networks (DANs) improve emotion recognition by employing a dual attention mechanism that emphasizes crucial temporal moments through temporal attention and significant frequency bands via channel attention. This approach enables the model to capture subtle emotional cues effectively. It effectively integrates self-attention to identify relationships within each modality alongside cross-modality attention that aligns features across different modalities. This approach prioritizes intra-modality and inter-modality characteristics, significantly enhancing results by capturing complex dependencies between modalities while preserving crucial modality-specific information. Additionally, the DANs improve contextual understanding, which is vital for recognizing emotions based on intonation, stress patterns, and prosody. Finally, the visualization of attention maps in DANs provides valuable insights into which elements of the speech signal contribute most to emotion recognition, improving model interpretability and refining recognition systems. Thus, the DANs are promising for advancing SER capabilities. Chen and Huang (2021) proposed that the dual attention-BiLSTM model leverages the strengths of raw audio signals by simultaneously extracting log Mel-spectrograms and MFCCs from the audio. Zaidi et al. (2024) employed DANs comprising Graph attention networks (GATs) and Co-attention to enhance the performance of their multimodal dual attention transformer (MDAT) framework for cross-language multimodal SER.

6.3.4. Graph attention networks

GATs (Veličković et al., 2018) use an attention mechanism to assess the importance of each neighboring node to the node under consideration. This allows information to be aggregated based on each neighbor of importance. The more critical an adjacent node is, the greater its influence on the information of the target node. The general function of GATs is visualized in Eq. (13).

$$m_u^{(l)} = W_m^{(l)} h_u^{(l-1)}, \tag{12}$$

where $W_m^{(l)} \in \mathbb{R}^{d' \times d}$ is the weight matrix that the model will learn.

$$h_u^{(l)} = \sigma\left(\sum_{v \in \mathcal{N}(u) \cup \{u\}} \alpha_{uv} m_v^{(l)}\right), \tag{13}$$

where σ is a nonlinear activation function, and α_{uv} represents the importance of node v to node u. α_{uv} usually called the attention coefficient, defined as in Eq. (15).

$$e_{uv}^{(l)} = F(m_u^{(l)}, m_v^{(l)}). \tag{14}$$

In this function, $F(\cdot)$ is a function that returns the similarity between two input vectors. Some common functions for $F(\cdot)$ are dot product attention (Lovisotto et al., 2022), end-to-end attention (Bahdanau et al., 2016), and Luong attention (Luong, 2015).

$$\alpha_{uv}^{(l)} = \frac{\exp(e_{uv}^{(l)})}{\sum_{k \in \mathcal{N}(u) \cup \{u\}} \exp(e_{uk}^{(l)})}. \tag{15}$$

An example of GATs is a study of Fu et al. (2024). The author proposed a multimodal fusion dialogue GATs (MM DialogueGATs). MM DialogueGAT is proposed to enable in-depth interactions across multiple modalities. The model employs GATs, which introduce an attention mechanism to adjust the importance of neighboring nodes dynamically. This allows each node to prioritize significant neighbors during updates. Using GNNs, MM DialogueGAT captures cross-dialogue jump information, enhancing the understanding of local and global context. This approach addresses complex dependencies within dialogue sequences, improving emotion recognition accuracy. Extensive experiments on multiple datasets demonstrate the model's effectiveness and superiority. Li et al. (2023c) proposed a novel multimodal fusion method called the graph network-based multimodal fusion technique (GraphMFT). In this approach, each conversation is represented by three heterogeneous graphs: visual-acoustic, visual-textual, and acoustic-textual. Each graph incorporates data from only two modalities to reduce the complexity of multimodal fusion. Additionally, they enhanced the vanilla GAT to address the over-smoothing problem of GNNs, drawing inspiration from residual GCN (ResGCN). GraphMFT was subsequently evaluated on two benchmark datasets commonly used by most ERC models.

GATs can model the varying strengths of correlations between nodes in the graph, thereby determining the amount of information that neighboring nodes can contribute to the target node. This flexibility allows multiple approaches to calculate the importance between nodes, enhancing GATs' generalization ability. As a result, GATs can be applied effectively to various tasks across multiple domains, including multimodal feature fusion for SER. In such applications, GATs can help prioritize and integrate features from different modalities, enabling more robust and accurate emotion recognition.

6.4. Graph-based fusion

GNNs were first introduced by Gori et al. (2005) and improved by Scarselli et al. (2009). Their ability to effectively represent graph structures while maintaining computational complexity has made them increasingly popular across various domains. Enhanced GNNs-based models, such as graph convolutional networks (GCNs), GATs, GraphSAGE, and gated GNNs (GGNNs), offer specialized methods for aggregating and processing graph data. GCNs, for instance, extend classical CNNs to work directly on graphs, which is highly effective for capturing local structures. GraphSAGE, on the other hand, introduces an inductive learning approach that enables GNNs to generalize to unseen nodes, making it ideal for dynamic or evolving graph structures. GGNNs employ GRUs to model sequential dependencies in graph-based data, allowing for more flexible information aggregation.

In SER, graph-based fusion utilizes graph structures to represent relationships among data points (nodes), such as acoustic and textual features. Fig. 16 illustrates the general workflow of GNNs, starting with input graph modeling. In this step, raw data is transformed into a structured graph format. Consider a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{v_i \mid i = 1, \dots, |\mathcal{V}|\}$ denotes the set of vertices, and $\mathcal{E} = \{e_{ij} \mid i, j \in \mathcal{V}\}$ represents the set of edges connecting the vertices. To incorporate graphs into deep learning models, the node feature matrix $\mathbf{X}$ and the adjacency matrix $\mathbf{A}$ can be computed as described in Eqs. (16) and (17).

$$\mathbf{X} = \left\{x_i \in \mathbb{R}^d\right\}, i = \overline{1, |V|}, \tag{16}$$

where x_i represents the feature vector of node v_i, with d being the dimensionality of the feature space. The node feature matrix $\mathbf{X}$ captures the intrinsic properties of each node.

$$A_{ij} = \begin{cases} 1 & \text{if there is a directed edge from } j \text{ to } i \\ 0 & \text{otherwise,} \end{cases} \tag{17}$$

where A_{ij} is an element of the adjacency matrix $\mathbf{A}$. It represents the connectivity between nodes v_j and v_i. Specifically, $A_{ij} = 1$ indicates the existence of a directed edge from node v_j (source) to node v_i (target), signifying a relationship or interaction between them. Conversely, $A_{ij} = 0$ means no such edge exists. The adjacency matrix $\mathbf{A}$, being of size $|V| \times |V|$, serves as a compact representation of the graph's structure and is fundamental for propagating information among nodes in GNNs.

Each node in the graph represents a fundamental unit of information derived from the input data, such as overlapping audio frames for audio or words, phrases, or sentences for text. The edges connecting these nodes signify the relationships or dependencies, reflecting temporal continuity in audio or semantic relevance in text. In an audio-based graph, nodes represent overlapping audio frames, with edges illustrating their transitions or similarities. Similarly, nodes may represent words or phrases in a textual graph, while the edges denote syntactic or semantic connections. Within the GNNs layer, node features undergo an initial transformation through a linear projection, followed by a

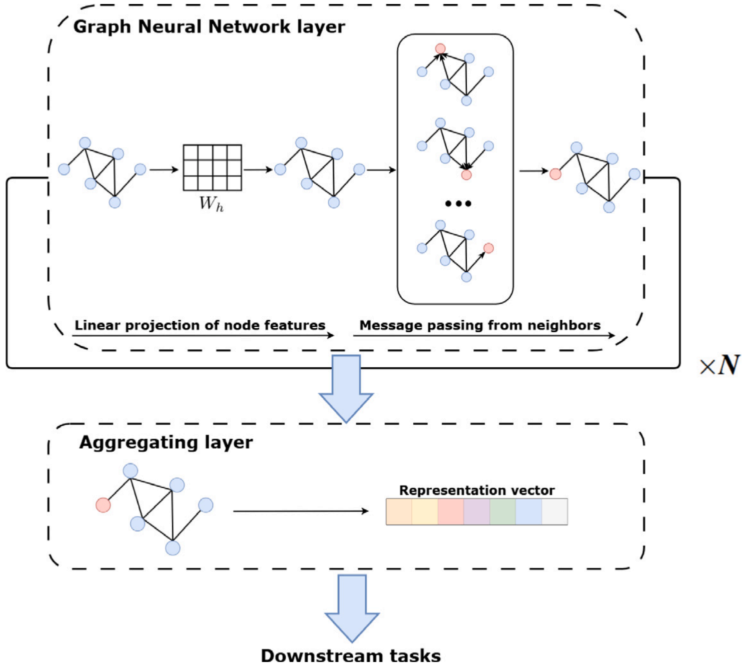


Fig. 16. The general architecture of GNNs.

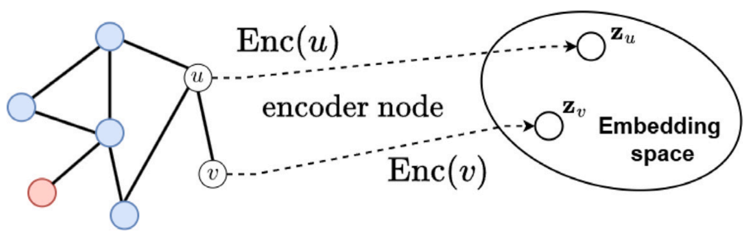


Fig. 17. General idea of node embedding.

message-passing step, where nodes aggregate information from their neighboring nodes to refine their representations. This process is reiterated across multiple layers to capture higher-order dependencies. The aggregation layer then combines the node representations into a graph-level representation vector using pooling methods such as sum, mean, or max pooling. Ultimately, this vector predicts the target output for downstream tasks, such as emotion classification or regression.

Node embedding is critical for efficiently representing nodes in a low-dimensional embedding space. This representation captures the essential features and relationships of nodes in the graph. Fig. 17 visualizes the general node embedding, in which $\mathrm{Enc}(u)$ is the encoder function for node u. There are three steps for node embedding. In the first step, we define an encoder that maps each node u in the graph to its corresponding embedding vector $\mathbf{z}_u \in \mathbb{R}^d$, which is visualized in Eq. (18).

$$\mathrm{Enc}(u) = \mathbf{z}_u. \tag{18}$$

To establish the similarity measure, node relationships are modeled using a similarity function that quantifies the relationship between nodes u and v. For instance, the similarity can be defined in Eq. (19).

$$\mathrm{Similarity}(u, v) = \frac{\mathbf{z}_u^\top \mathbf{z}_v}{\|\mathbf{z}_u\|\|\mathbf{z}_v\|}. \tag{19}$$

The objective is to optimize the embeddings so that the similarity between nodes u and v aligns with their actual relationships. This is achieved by defining a loss function $\mathcal{L}$ to minimize the difference, expressed as Eq. (20).

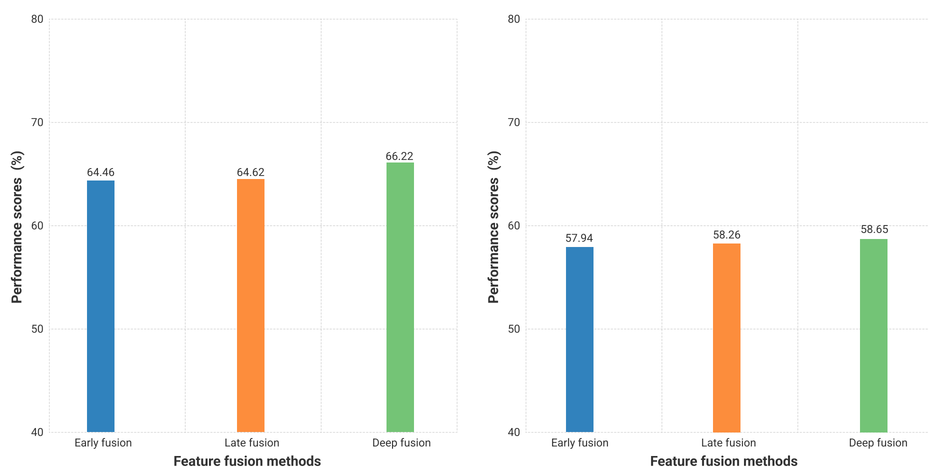
$$\mathcal{L} = -\log\left(\mathrm{Similarity}(u, v)\right). \tag{20}$$

Finally, the encoder parameters are adjusted iteratively to minimize the loss function $\mathcal{L}$ and improve the quality of the embeddings. As $\mathcal{L}$ approaches zero, the learned embeddings effectively preserve the underlying relationships within the graph.

For instance, Liu et al. (2022a) reformulated the SER problem as a graph classification task by converting variable-length utterances into graph structures, eliminating the need for padding or truncation. In their approach, each frame (a short, windowed segment within an utterance) is represented as a node in the graph. In contrast, Pentari et al. (2024) leveraged well-established graph-based features derived from two distinct adjacency matrices. Their method effectively captures both structural and statistical properties of speech signals, providing a more robust feature representation. Meng et al. (2024) introduced multivariate message-passing GCNs (MMPGCNs), integrating semantic dialogue context, speaker relationships, and multimodal heterogeneity to enhance emotion tracking across conversations. Yuan et al. (2023) achieved the highest accuracy among these approaches, proposing the relational bilevel aggregation GCNs (RBA-GCNs), which effectively model long-range contextual dependencies and inter-modality interactions within a single-layer architecture. Their method demonstrates the benefits of their novel graph generation module (GGM) and similarity-based cluster building module (SCBM). These components enhance

Table 10
Performance comparison highlighting the impact of graph-based fusion.

Paper	Feature fusion methods	Dataset	Graph-based fusion	Performance	
				WA (%)	UA (%)
Fu et al. (2024)	Deep fusion	IEMOCAP	✗	77.12	76.88
			✓	81.46	81.99
Fan et al. (2024)	Deep fusion	IEMOCAP	✗	66.67	66.61
			✓	69.34	69.57
		MELD	✗	64.67	65.41
			✓	66.05	67.13
Pentari et al. (2024)	Early fusion	Emo-DB	✗	–	59.89
			✓	–	77.80
		DEMoS	✗	–	72.76
			✓	–	79.10
		AESDD	✗	–	57.10
			✓	–	70.00
Yuan et al. (2023)	Deep fusion	IEMOCAP	✗	70.57	–
			✓	71.43	–
		MELD	✗	57.33	–
			✓	62.67	–



(a) On the IEMOCAP dataset. (b) On the MELD dataset.

Fig. 18. Accuracy comparison of graph-based fusion methods. This figure is generated based on the experimental results reported by Hu et al. (2021).

inter-class relationships through advanced similarity metrics, effectively reducing redundancy at target nodes and improving multimodal fusion in SER.

Based on the performance comparison shown in Table 10, it is evident that fusion methods augmented with graph-based mechanisms consistently surpass their non-graph counterparts across diverse datasets and fusion schemes. This performance gain is observed in both early and deep fusion approaches, where the integration of graph modeling results in substantial improvements. The performance highlights the strength of graph modeling in capturing long-range dependencies, inter-modal interactions, and contextual relationships. By structuring data as graphs, models gain more robust and expressive feature representations, resilient to noise and missing data. Graph-enhanced fusion thus offers a powerful, scalable, and model-agnostic solution that significantly boosts the effectiveness of multimodal SER.

More specifically, Hu et al. (2021) explicitly evaluated different integration points using their GCN architecture. They compared early fusion, late fusion, and their proposed deep fusion. As depicted in Fig. 18, the deep fusion approach consistently outperformed the others on both benchmark datasets. On the IEMOCAP, deep fusion achieved a UA of 66.22%, compared to 64.62% for late fusion and 64.46% for early fusion. On the MELD dataset, deep fusion led with a UA of 58.65%, surpassing late fusion (58.26%) and early fusion (57.94%). This study provides strong evidence that, even when leveraging graph structures, integrating multimodal information interactively at deeper representational layers within the graph network offers significant accuracy benefits. This accuracy advantage underscores the importance of deep interaction mechanisms within the GCN itself for effective multimodal graph-based fusion in SER.

Despite these accuracy gains, the computational complexity introduced by graph-based methods remains an underexplored dimension in the literature. The construction and processing of graphs can significantly increase memory usage, training time, and inference latency. While models demonstrate clear benefits in accuracy, the trade-off regarding resource consumption is seldom reported or analyzed in depth. Therefore, future research should adopt a more holistic evaluation approach that includes efficiency metrics such as FLOPs, parameter count, and runtime performance, ensuring that the practical deployment of these models remains feasible for real-time or resource-constrained environments.

7. Open challenges and future works

7.1. Insights and analysis

7.1.1. RQ 1: What publicly accessible multimodal datasets are commonly used for multimodal fusion in SER?

The standard datasets for multimodal fusion in SER studies include IEMOCAP (39.8%), RAVDESS (11.4%), MELD (11.4%), and EMO-DB (10.2%). These datasets are favored for their diversity of emotional expressions, enabling researchers to evaluate models across a wide range of emotions, which is essential for real-world applications. They typically combine audio, video, and text data, facilitating research on how modality integration enhances emotion recognition. IEMOCAP and MELD offer controlled and spontaneous emotional expressions, balancing training consistency with natural emotion capture. As benchmark datasets, they foster consistent comparisons in results, and their public availability encourages collaboration and advances SER technology.

7.1.2. RQ 2: Which fusion methods can most effectively integrate multimodal data for SER, and how do they impact performance?

Each fusion method offers unique strengths, but hybrid and graph-based fusion is particularly effective for achieving high accuracy in multimodal SER tasks. The hybrid fusion method combines early, deep, and late fusion methods, capturing granular and overarching emotional cues and making them versatile for complex SER tasks. On the other hand, graph-based fusion models represent each modality as a node in a graph with edges indicating the inter-modal relationships, allowing the model to prioritize the most relevant connections dynamically. Both methods integrate diverse emotional signals, with graph-based approaches being beneficial for context-dependent emotions in conversational data. Attention mechanisms, such as cross-modality and MHA, substantially enhance performance by focusing on the most critical emotional features within and across various modalities. This advancement dramatically improves the interpretability and accuracy of SER systems. Consequently, hybrid and graph-based fusion models are exceptionally well-suited for real-world SER applications that require a deep understanding of emotions.

7.1.3. RQ 3: How do fusion techniques enhance the accuracy and interpretability of multimodal SER models?

Multimodal SER models achieve higher performance when integrating advanced fusion techniques. Graph-based fusion has demonstrated the most effective results among the primary approaches, leveraging GNNs and GATs to model inter-modal relationships, preserving dependencies, and enhancing context awareness. Attention mechanisms, particularly MHA and cross-modality attention, significantly improve SER performance when combined with a deep or hybrid fusion method. These mechanisms prioritize essential emotional cues across modalities, ensuring that only the most relevant features contribute to classification. By leveraging graph-based fusion, attention mechanisms, and data translation, the multimodal SER models achieve greater accuracy and robustness, setting new benchmarks in emotion recognition tasks.

7.1.4. RQ 4: What are the main challenges associated with multimodal fusion in SER, and how can they be addressed?

Data alignment and synchronization are crucial, as each modality captures emotions differently, requiring advanced alignment and temporal attention mechanisms to match cues accurately. Modality-specific noise and variability are addressed through denoising and attention-based filtering to focus on relevant signals. Heterogeneity across modalities adds complexity, as each offers unique emotional information, cross-modal attention, and hybrid fusion to help unify these differences. High dimensionality and computational demands can also risk overfitting and increase processing needs, so techniques like dimensionality reduction and model optimization are exploited to streamline fusion. Specifically, attention mechanisms play a pivotal role in improving the interpretability of multimodal SER models, making how each feature contributes to emotion detection more transparent. Together, these solutions help create more reliable and accurate SER models.

7.2. Open challenges and future directions

This survey highlights several open challenges and outlines directions for future research in SER. One major challenge is integrating recent advancements, particularly transformer-based architectures, which effectively capture complex dependencies across multiple modalities. Although pre-trained models and multimodal transformers have demonstrated the potential for enhancing SER performance, further incorporating these innovations is crucial. Another significant challenge is the limitations of current datasets, which often fail to encompass the full spectrum of emotional expressions, especially across different cultural contexts. Urgent efforts are needed to address these shortcomings and assess the scalability of SER systems in resource-constrained environments for broader applicability. Additionally, enhancing the robustness and adaptability of SER technologies will require including diverse datasets and state-of-the-art techniques to bridge existing gaps. Future research will concentrate on developing advanced architectures, refining fusion techniques, and enhancing real-time efficiency to produce more robust and adaptable SER systems, making them suitable for a wide range of real-world applications. Several directions for further research include:

7.2.1. Data limitations

Most multimodal SER datasets contain acted or spontaneous emotions recorded in controlled environments, lacking the variability of real-world speech. This limits generalization, as models trained on clean, balanced data struggle with noise, overlapping speech, and contextual diversity in practical applications. A key issue is the absence of noise, which suppresses implicit emotional cues. Real-world interactions often include background disturbances that affect speech quality, making models trained on noise-free datasets less robust. Additionally, data imbalance skews recognition accuracy, as some emotions (e.g., happiness, neutral) are overrepresented while others (e.g., anger, fear) are underrepresented, leading to biased predictions.

o address these challenges, expanding dataset collection to real-world scenarios, applying data augmentation such as noise injection (Saon et al., 2019; Feng et al., 2022), speaker variations (Qian et al., 2022), and leveraging self-supervised learning (SSL) can improve model robustness (Han et al., 2025; Mohamed et al., 2022). Domain adaptation and balanced training strategies based on resampling (Furundzic et al., 2017) and adaptive loss functions (Kolbæk et al., 2020) can further enhance generalization, making SER models more effective for real-world deployment.

7.2.2. Data robustness

Multimodal SER faces critical challenges due to synchronization issues and noise interference. Misalignment between modalities can lead to inconsistent temporal correspondence, weakening the model's ability to associate emotional cues correctly. Additionally, real-world audio recordings are often contaminated with background noise, reverberations, and speech overlaps, all of which can degrade signal quality. These factors make building reliable and generalizable multimodal SER systems difficult, especially in unconstrained environments.

Researchers have developed explicit and implicit alignment strategies to overcome synchronization challenges. Explicit alignment methods such as Dynamic Time Warping (DTW) (Müller, 2007) and Deep Canonical Correlation Analysis (DCCA) (Andrew et al., 2013) establish structured correspondences between modalities, ensuring precise mapping of speech and textual features. Meanwhile, implicit alignment approaches, including graph-based cross-modal alignment (Song et al., 2023) and attention-based alignment (Tu et al., 2023), leverage learned relationships within latent spaces to synchronize modalities during training dynamically. To address noise in speech emotion data, pre-processing techniques such as spectral subtraction, voice activity detection (VAD) (Graf et al., 2015), and denoising autoencoders (DAE) (Gupta et al., 2022) help filter out irrelevant noise and enhance signal quality. Furthermore, model-based strategies such as adversarial training, self-supervised learning, and domain adaptation enhance generalization across diverse recording conditions, thereby mitigating the impact of noisy environments. Integrating these alignment and denoising methods ensures robust multimodal representations, ultimately improving emotion recognition performance in real-world scenarios.

7.2.3. Addressing cross-cultural variability in multimodal SER

One of the key challenges in a multimodal SER dataset is its inability to generalize across different cultures. Emotional expression varies significantly between languages, social norms, and cultural contexts, making it difficult for SER models trained on one dataset to generalize to another. Research should incorporate culturally diverse datasets to improve model adaptability. Moreover, transfer learning can fine-tune SER models for specific cultural groups. Researchers are also applying alignment techniques to identify universal emotional patterns across languages and cultures. These techniques help reduce biases caused by variations in intonation, language, and emotional expression, thereby enhancing the generalization capability of SER models (Pham et al., 2023; Vekkot and Gupta, 2022).

7.2.4. Optimizing for real-time processing and computational complexity

Real-time processing is critical for SER applications, especially in live interactive environments. Model optimization techniques such as pruning, quantization, and caching are essential to achieve fast, low-latency inference. These strategies significantly reduce model size and computational demands, making multimodal SER systems practical for deployment on resource-constrained devices such as mobile phones and embedded systems.

In high-infrastructure environments, latency can be minimized through hardware acceleration using Graphics Processing Units (GPUs), Tensor Processing Units (TPUs), or cloud-based inference, allowing complex deep learning models to deliver rapid emotion recognition (Chan et al., 2023). However, such hardware is often unavailable in low-infrastructure settings, where real-time audio processing, feature extraction, and classification must operate on Central Processing Units (CPUs) or edge devices with limited compute resources (Lu et al., 2023). In these scenarios, achieving low-latency inference becomes more challenging, and inefficient models can lead to delayed or inaccurate emotion predictions.

Another practical challenge in real-time deployment is the occasional unavailability of one or more modalities due to sensor limitations, network latency, or user behavior (Chen et al., 2025; Menon et al., 2025). This missing modality problem can compromise model reliability unless addressed through robust, modality-agnostic fusion strategies that enable the system to function effectively with partial inputs. Therefore, further optimization is necessary to adapt SER models for deployment under constrained and incomplete conditions.

Transfer learning further enhances efficiency by leveraging pre-trained embeddings from large-scale speech and text models, improving feature extraction while minimizing labeled data requirements (Yan et al., 2024b; Triantafyllopoulos and Schuller, 2021). Model compression techniques, such as knowledge distillation, ensure scalability by transferring knowledge from large, high-performance models to smaller, lightweight models without sacrificing accuracy (Caldeira et al., 2025; Nguyen et al., 2024b; Zhu et al., 2025). Pruning eliminates redundant parameters (Mohammadi et al., 2023), reducing model size while retaining critical features, whereas quantization lowers numerical precision in deep networks, optimizing memory usage and enabling faster inference (Liang et al., 2021).

To handle missing modalities in real-time multimodal emotion recognition, recent methods focus on either recovering the missing information or improving model robustness to incomplete inputs. Cross-modality imagination (Zhao et al., 2021) predicts missing representations using available modalities, often supported by techniques like Central Moment Discrepancy (Zuo et al., 2023) and Contrastive Learning (Liu et al., 2024a) to learn modality-invariant features. Score-based diffusion models (Wang et al., 2023b) offer another solution by mapping noise to the target modality distribution. Additionally, training-time data augmentation can improve robustness, such as randomly ablating modalities (Goncalves and Busso, 2022). However, these approaches, especially diffusion models, can incur high computational overhead. Thus, they often need to be paired with lightweight optimization strategies like pruning, quantization, or knowledge distillation to ensure real-time feasibility. By integrating these cutting-edge optimization techniques, multimodal SER systems can deliver high-performance emotion recognition while remaining efficient, scalable, and suitable for real-world deployment.

7.2.5. Lack of standardized evaluation for computational costs in multimodal SER

Despite notable advancements in improving classification accuracy through various feature fusion methods, computational complexity remains insufficiently addressed in multimodal SER. Most existing studies prioritize performance metrics like WA and UA while neglecting to report or compare critical computational indicators such as FLOPs, parameter count, model size, and inference latency. This oversight limits our understanding of performance gains and resource consumption trade-offs. Given the practical need to deploy SER systems on real-time or resource-constrained platforms, the future direction must move toward a more holistic evaluation framework. This includes incorporating standardized efficiency metrics and establishing common benchmarks for comparing fusion methods not only in terms of accuracy but also in terms of computational cost. Bridging this gap will promote the development of scalable, deployable, and energy-efficient multimodal SER systems.

8. Conclusion

This survey delivers a comprehensive and in-depth analysis of multimodal fusion techniques in emotion recognition, demonstrating that integrating complementary modalities is essential for achieving superior classification performance. We systematically evaluate early, late, deep, and hybrid fusion methods based on groundbreaking research from 2020 to 2024, identifying their strengths, limitations, and impact on model accuracy. Our findings highlight that advanced fusion techniques, such as data representation, data translation, attention mechanisms, and particularly graph-based fusion, are pivotal in enhancing multimodal SER. These innovations improve model robustness and establish new benchmarks for handling complex, real-world emotional data. Beyond technical advancements, this study critically addresses persistent challenges such as data imbalance, cross-cultural variability in SER, the need for real-time processing, and computational complexity. We also examine state-of-the-art model compression techniques to streamline deployment in practical, real-world applications. This survey offers valuable insights to guide future research and industrial applications by bridging theoretical innovations with practical implementation. Multimodal SER is poised to transform key sectors, including healthcare, education, and virtual assistants. Our work lays a solid foundation for developing more intelligent, scalable, and emotion-aware artificial intelligence systems.

Abbreviations

Abbreviation	Definition
1D-CNNs	1-Dimensional convolutional neural networks
ACRNN	Auto-conditioned recurrent network
BERT	Bidirectional encoder representations from transformers
BiGRU	Bidirectional gated recurrent unit
BiLSTM	Bidirectional long short-term memory
CBAM	Convolutional block attention module
CMFA	Cross-modal fusion attention
CNNs	Convolutional neural networks
CPUs	Central processing units
DANs	Dual attention networks
DBMER	Deep learning-based multi-learning model for emotion recognition
DL	Deep learning
DNN	Deep neural network
eGeMAPS	Extended Geneva Minimalistic Acoustic Parameter Set
FLOPs	Floating point operations per second
GANs	Generative adversarial networks
GATs	Graph attention networks
GCNs	Graph convolutional networks
GGM	Graph generation module
GGNN	Gate graph neural network
GeMAPS	Geneva minimalistic acoustic parameter set
GNNs	Graph neural networks
GPUs	Graphics processing units
GPT	Generative pre-trained transformer
GRU	Gated recurrent unit
HSFs	High-level statistical functions
ISSA	Improved variational sparrow search algorithm
LGCN	Learnable graph convolutional network
LLDs	Low-level descriptors
LSTM	Long short-term memory
MDAT	Multimodal dual attention transformer
MFCC	Mel frequency cepstral coefficient
MHA	Multi-head self-attention
ML	Machine learning
MLP	Multi-layer perceptron
MM DialogueGATs	Multimodal fusion dialogue graph attention network
MMPGCN	Multivariate message passing GCN
MSA	Multi-head self-attention
NLP	Natural Language Processing
PAD	Pleasure-arousal-dominance model
PRISMA	Preferred reporting items for systematic reviews and meta-analyses
RBA-GCNs	Relational bilevel aggregation GCNs
ResGCN	Residual graph convolutional neural network
ResNet	Residual network
RNN	Recurrent neural networks
RoBERTa	Robustly optimized BERT pretraining approach
RQ	Research questions
SAM	Self-attention model
SCBM	Similarity-based cluster building module
SER	Speech emotion recognition
SVM	Support vector machine
TFCNNs	Temporal-frequency convolutional neural networks
TFN	Tensor fusion network
TPU	Tensor processing unit
UA	Unweighted accuracy
VA	Valence-arousal model
WA	Weighted accuracy
ZCR	Zero crossing rate

CRediT authorship contribution statement

Nhut Minh Nguyen: Writing – original draft, Visualization, Software, Methodology, Conceptualization. **Thanh Trung Nguyen:** Visualization, Software, Methodology. **Phuong-Nam Tran:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Chee Peng Lim:** Writing – review & editing, Writing – original draft, Supervision. **Nhat Truong Pham:** Writing – review & editing, Writing – original draft, Validation, Supervision, Conceptualization. **Duc Ngoc Minh Dang:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This research is funded by FPT University under grant number DHFPT/2025/10.

This work was done during Nhat Truong Pham's research visit to the AiTA Lab, FPT University, Vietnam.

Data availability

The data used in this study, including datasets such as IEMOCAP, MELD, RAVDESS, and others mentioned, are publicly available from their respective sources. The manuscript provides further details on dataset access and pre-processing methods. Our up-to-date repository for a comprehensive study on multimodal fusion in speech emotion recognition can be publicly accessible at https://github.com/aita-lab/awesome-multimodal-fusion-emotion-recognition/.

References

Abdusalomov, A., Kutlimuratov, A., Nasimov, R., Whangbo, T.K., 2023. Improved speech emotion recognition focusing on high-level data representations and swift feature extraction calculation. Comput. Mater. Contin. 77 (3), 2915–2933. http://dx.doi.org/10.32604/cmc.2023.044466.

Ai, W., Zhang, F., Shou, Y., Meng, T., Chen, H., Li, K., 2025. Revisiting multimodal emotion recognition in conversation from the perspective of graph spectrum. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 39, (11), pp. 11418–11426. http://dx.doi.org/10.1609/aaai.v39i11.33242, URL: https://ojs.aaai.org/index.php/AAAI/article/view/33242.

Andrew, G., Arora, R., Bilmes, J., Livescu, K., 2013. Deep canonical correlation analysis. In: Dasgupta, S., McAllester, D. (Eds.), Proceedings of the 30th International Conference on Machine Learning. In: Proceedings of Machine Learning Research, vol. 28, (3), PMLR, Atlanta, Georgia, USA, pp. 1247–1255, URL: https://proceedings.mlr.press/v28/andrew13.html.

Asai, A., Min, S., Zhong, Z., Chen, D., 2023. Retrieval-based language models and applications. In: Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 6: Tutorial Abstracts). pp. 41–46. http://dx.doi.org/10.18653/v1/2023.acl-tutorials.6.

Atila, O., Şengür, A., 2021. Attention guided 3D CNN-LSTM model for accurate speech based emotion recognition. Appl. Acoust. 182, 108260. http://dx.doi.org/10.1016/j.apacoust.2021.108260.

Atmaja, B.T., Sasou, A., Akagi, M., 2022. Survey on bimodal speech emotion recognition from acoustic and linguistic information fusion. Speech Commun. 140, 11–28. http://dx.doi.org/10.1016/j.specom.2022.03.002.

Baevski, A., Zhou, Y., Mohamed, A., Auli, M., 2020. wav2vec 2.0: A framework for self-supervised learning of speech representations. In: Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., Lin, H. (Eds.), Adv. Neural Inf. Process. Syst. 33, 12449–12460, URL: https://proceedings.neurips.cc/paper_files/paper/2020/file/92d1e1eb1cd6f9fba3227870bb6d7f07-Paper.pdf.

Bahdanau, D., Chorowski, J., Serdyuk, D., Brakel, P., Bengio, Y., 2016. End-to-end attention-based large vocabulary speech recognition. In: 2016 IEEE International Conference on Acoustics, Speech and Signal Processing. ICASSP, pp. 4945–4949. http://dx.doi.org/10.1109/ICASSP.2016.7472618.

Baltrušaitis, T., Ahuja, C., Morency, L.P., 2019. Multimodal machine learning: A survey and taxonomy. IEEE Trans. Pattern Anal. Mach. Intell. 41 (2), 423–443. http://dx.doi.org/10.1109/TPAMI.2018.2798607.

Bengio, Y., Courville, A., Vincent, P., 2013. Representation learning: A review and new perspectives. IEEE Trans. Pattern Anal. Mach. Intell. 35 (8), 1798–1828. http://dx.doi.org/10.1109/TPAMI.2013.50.

Burkhardt, F., Paeschke, A., Rolfes, M., Sendlmeier, W.F., Weiss, B., et al., 2005. A database of German emotional speech. In: Interspeech, vol. 5, pp. 1517–1520.

Busso, C., Bulut, M., Lee, C.C., Kazemzadeh, A., Mower, E., Kim, S., Chang, J.N., Lee, S., Narayanan, S.S., 2008. IEMOCAP: Interactive emotional dyadic motion capture database. Lang. Resour. Eval. 42, 335–359. http://dx.doi.org/10.1007/s10579-008-9076-6.

Cairong, Z., Xinran, Z., Cheng, Z., Li, Z., 2016. A novel DBN feature fusion model for cross-corpus speech emotion recognition. J. Electr. Comput. Eng. 2016 (1), 7437860. http://dx.doi.org/10.1155/2016/7437860.

Caldeira, E., Neto, P.C., Huber, M., Damer, N., Sequeira, A.F., 2025. Model compression techniques in biometrics applications: A survey. Inf. Fusion 114, 102657. http://dx.doi.org/10.1016/j.inffus.2024.102657, URL: https://www.sciencedirect.com/science/article/pii/S1566253524004354.

Chan, K.Y., Abu-Salih, B., Qaddoura, R., Al-Zoubi, A.M., Palade, V., Pham, D.S., Ser, J.D., Muhammad, K., 2023. Deep neural networks in the cloud: Review, applications, challenges and research directions. Neurocomputing 545, 126327. http://dx.doi.org/10.1016/j.neucom.2023.126327, URL: https://www.sciencedirect.com/science/article/pii/S0925231223004502.

Chen, G., Chen, L., Jiao, S., Tan, L., 2025. A low heterogeneity missing modality recovery learning for speech–visual emotion recognition. Expert Syst. Appl. 266, 126070. http://dx.doi.org/10.1016/j.eswa.2024.126070, URL: https://www.sciencedirect.com/science/article/pii/S0957417424029373.

Chen, L., Feng, G., Leong, C.W., Lehman, B., Martin-Raugh, M., Kell, H., Lee, C.M., Yoon, S.Y., 2016. Automated scoring of interview videos using Doc2Vec multimodal feature extraction paradigm. In: Proceedings of the 18th ACM International Conference on Multimodal Interaction. pp. 161–168. http://dx.doi.org/10.1145/2993148.2993203.

Chen, Y., Hou, W., Cheng, X., Li, S., 2018b. Joint learning for emotion classification and emotion cause detection. In: Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing. pp. 646–651. http://dx.doi.org/10.18653/v1/D18-1066.

Chen, Q., Huang, G., 2021. A novel dual attention-based BLSTM with hybrid features in speech emotion recognition. Eng. Appl. Artif. Intell. 102, 104277. http://dx.doi.org/10.1016/j.engappai.2021.104277.

Chen, Z., Lin, M., Wang, Z., Zheng, Q., Liu, C., 2023. Spatio-temporal representation learning enhanced speech emotion recognition with multi-head attention mechanisms. Knowl.-Based Syst. 281, 111077. http://dx.doi.org/10.1016/j.knosys.2023.111077.

Chen, H., Zendehdel, N., Leu, M.C., Yin, Z., 2023a. Fine-grained activity classification in assembly based on multi-visual modalities. J. Intell. Manuf. 35 (5), 2215–2233. http://dx.doi.org/10.1007/s10845-023-02152-x.

Costantini, G., Iaderola, I., Paoloni, A., Todisco, M., 2014. EMOVO corpus: an Italian emotional speech database. In: Calzolari, N., Choukri, K., Declerck, T., Loftsson, H., Maegaard, B., Mariani, J., Moreno, A., Odijk, J., Piperidis, S. (Eds.), Proceedings of the Ninth International Conference on Language Resources and Evaluation. LREC'14, European Language Resources Association (ELRA), Reykjavik, Iceland, pp. 3501–3504, URL: http://www.lrec-conf.org/proceedings/lrec2014/pdf/591_Paper.pdf.

Desai, K., Nickel, M., Rajpurohit, T., Johnson, J., Vedantam, S.R., 2023. Hyperbolic image-text representations. In: International Conference on Machine Learning. PMLR, pp. 7694–7731.

Devlin, J., Chang, M.W., Lee, K., Toutanova, K., 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In: Burstein, J., Doran, C., Solorio, T. (Eds.), Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers). Association for Computational Linguistics, Minneapolis, Minnesota, pp. 4171–4186. http://dx.doi.org/10.18653/v1/N19-1423, URL: https://aclanthology.org/N19-1423/.

D'Mello, S., Kory, J., 2012. Consistent but modest: a meta-analysis on unimodal and multimodal affect detection accuracies from 30 studies. In: Proceedings of the 14th ACM International Conference on Multimodal Interaction. pp. 31–38. http://dx.doi.org/10.1145/2388676.2388686.

Ezzameli, K., Mahersia, H., 2023. Emotion recognition from unimodal to multimodal analysis: A review. Inf. Fusion 99, 101847. http://dx.doi.org/10.1016/j.inffus.2023.101847.

Fan, C., Lin, J., Mao, R., Cambria, E., 2024. Fusing pairwise modalities for emotion recognition in conversations. Inf. Fusion 106, 102306. http://dx.doi.org/10.1016/j.inffus.2024.102306, URL: https://www.sciencedirect.com/science/article/pii/S1566253524000848.

Fei, H., Zhang, H., Wang, B., Liao, L., Liu, Q., Cambria, E., 2024. EmpathyEar: An open-source avatar multimodal empathetic chatbot. In: Cao, Y., Feng, Y., Xiong, D. (Eds.), Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations). Association for Computational Linguistics, Bangkok, Thailand, pp. 61–71. http://dx.doi.org/10.18653/v1/2024.acl-demos.7, URL: https://aclanthology.org/2024.acl-demos.7/.

Feng, T., Hashemi, H., Annavaram, M., Narayanan, S.S., 2022. Enhancing privacy through domain adaptive noise injection for speech emotion recognition. In: ICASSP 2022 - 2022 IEEE International Conference on Acoustics, Speech and Signal Processing. ICASSP, pp. 7702–7706. http://dx.doi.org/10.1109/ICASSP43922.2022.9747265.

Fu, R., Gai, X., Abdulhakim Al-Absi, A., Abdulhakim Al-Absi, M., Alam, M., Li, Y., Jiang, M., Wang, X., 2024. MM DialogueGAT—A fusion graph attention network for emotion recognition using multi-model system. IEEE Access 12, 150941–150952. http://dx.doi.org/10.1109/ACCESS.2024.3350156.

Furundzic, D., Stankovic, S., Jovicic, S., Punisic, S., Subotic, M., 2017. Distance based resampling of imbalanced classes: With an application example of speech quality assessment. Eng. Appl. Artif. Intell. 64, 440–461. http://dx.doi.org/10.1016/j.engappai.2017.07.001, URL: https://www.sciencedirect.com/science/article/pii/S0952197617301495.

Gangamohan, P., Kadiri, S.R., Yegnanarayana, B., 2016. Analysis of emotional speech—A review. In: Toward Robotic Socially Believable Behaving Systems-Volume I: Modeling Emotions. Springer, pp. 205–238. http://dx.doi.org/10.1007/978-3-319-31056-5_11.

Geetha, A., Mala, T., Priyanka, D., Uma, E., 2024. Multimodal emotion recognition with deep learning: Advancements, challenges, and future directions. Inf. Fusion 105, 102218. http://dx.doi.org/10.1016/j.inffus.2023.102218, URL: https://www.sciencedirect.com/science/article/pii/S1566253523005341.

George, S.M., Ilyas, P.M., 2024. A review on speech emotion recognition: a survey, recent advances, challenges, and the influence of noise. Neurocomputing 568, 127015. http://dx.doi.org/10.1016/j.neucom.2023.127015.

Goncalves, L., Busso, C., 2022. Robust audiovisual emotion recognition: Aligning modalities, capturing temporal information, and handling missing features. IEEE Trans. Affect. Comput. 13 (4), 2156–2170. http://dx.doi.org/10.1109/TAFFC.2022.3216993.

Gori, M., Monfardini, G., Scarselli, F., 2005. A new model for learning in graph domains. In: Proceedings. 2005 IEEE International Joint Conference on Neural Networks, 2005, vol. 2, pp. 729–734 vol. 2. http://dx.doi.org/10.1109/IJCNN.2005.1555942.

Górriz, J., Álvarez-Illán, I., Álvarez-Marquina, A., Arco, J., Atzmueller, M., Ballarini, F., Barakova, E., Bologna, G., Bonomini, P., Castellanos-Dominguez, G., Castillo-Barnes, D., Cho, S., Contreras, R., Cuadra, J., Domínguez, E., Domínguez-Mateos, F., Duro, R., Elizondo, D., Fernández-Caballero, A., Fernandez-Jover, E., Formoso, M., Gallego-Molina, N., Gamazo, J., González, J.G., Garcia-Rodriguez, J., Garre, C., Garrigós, J., Gómez-Rodellar, A., Gómez-Vilda, P., Graña, M., Guerrero-Rodriguez, B., Hendrikse, S., Jimenez-Mesa, C., Jodra-Chuan, M., Julian, V., Kotz, G., Kutt, K., Leming, M., de Lope, J., Macas, B., Marrero-Aguiar, V., Martinez, J., Martinez-Murcia, F., Martínez-Tomás, R., Mekyska, J., Nalepa, G., Novais, P., Orellana, D., Ortiz, A., Palacios-Alonso, D., Palma, J., Pereira, A., Pinacho-Davidson, P., Pinninghoff, M., Ponticorvo, M., Psarrou, A., Ramírez, J., Rincón, M., Rodellar-Biarge, V., Rodríguez-Rodríguez, I., Roelofsma, P., Santos, J., Salas-Gonzalez, D., Salcedo-Lagos, P., Segovia, F., Shoeibi, A., Silva, M., Simic, D., Suckling, J., Treur, J., Tsanas, A., Varela, R., Wang, S., Wang, W., Zhang, Y., Zhu, H., Zhu, Z., Ferrández-Vicente, J., 2023. Computational approaches to explainable artificial intelligence: Advances in theory, applications and trends. Inf. Fusion 100, 101945. http://dx.doi.org/10.1016/j.inffus.2023.101945, URL: https://www.sciencedirect.com/science/article/pii/S1566253523002610.

Graf, S., Herbig, T., Buck, M., Schmidt, G., 2015. Features for voice activity detection: a comparative analysis. EURASIP J. Adv. Signal Process. 2015, 1–15.

Guo, Y., Zhou, Y., Xiong, X., Jiang, X., Tian, H., Zhang, Q., 2023. A multi-feature fusion speech emotion recognition method based on frequency band division and improved residual network. IEEE Access 11, 86013–86024. http://dx.doi.org/10.1109/ACCESS.2023.3299822.

Gupta, C., Li, H., Goto, M., 2022. Deep learning approaches in topics of singing information processing. IEEE/ACM Trans. Audio, Speech, Lang. Process. 30, 2422–2451. http://dx.doi.org/10.1109/TASLP.2022.3190732.

Hallmen, T., Deuser, F., Oswald, N., André, E., 2024. Unimodal multi-task fusion for emotional mimicry intensity prediction. In: 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops. CVPRW, pp. 4657–4665. http://dx.doi.org/10.1109/CVPRW63382.2024.00468.

Han, J., Landini, F., Rohdin, J., Silnova, A., Diez, M., Burget, L., 2025. Leveraging self-supervised learning for speaker diarization. In: ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing. ICASSP, pp. 1–5. http://dx.doi.org/10.1109/ICASSP49660.2025.10889475.

Han, S., Leng, F., Jin, Z., 2021. Speech emotion recognition with a ResNet-CNN-transformer parallel neural network. In: 2021 International Conference on Communications, Information System and Computer Engineering. CISCE, pp. 803–807. http://dx.doi.org/10.1109/CISCE52179.2021.9445906.

Hazarika, D., Poria, S., Zadeh, A., Cambria, E., Morency, L.P., Zimmermann, R., 2018. Conversational memory network for emotion recognition in dyadic dialogue videos. In: Walker, M., Ji, H., Stent, A. (Eds.), Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers). Association for Computational Linguistics, New Orleans, Louisiana, pp. 2122–2132. http://dx.doi.org/10.18653/v1/N18-1193, URL: https://aclanthology.org/N18-1193/.

Hazarika, D., Zimmermann, R., Poria, S., 2020. MISA: Modality-invariant and -specific representations for multimodal sentiment analysis. In: Proceedings of the 28th ACM International Conference on Multimedia. MM '20, Association for Computing Machinery, New York, NY, USA, pp. 1122–1131. http://dx.doi.org/10.1145/3394171.3413678.

He, Y., Xiang, S., Kang, C., Wang, J., Pan, C., 2016. Cross-modal retrieval via deep and bidirectional representation learning. IEEE Trans. Multimed. 18 (7), 1363–1377. http://dx.doi.org/10.1109/TMM.2016.2558463.

Ho, N.H., Yang, H.J., Kim, S.H., Lee, G., 2020. Multimodal approach of speech emotion recognition using multi-level multi-head fusion attention-based recurrent neural network. IEEE Access 8, 61672–61686. http://dx.doi.org/10.1109/ACCESS.2020.2984368.

Hong, S., Kang, H., Cho, H., 2024. Cross-modal dynamic transfer learning for multimodal emotion recognition. IEEE Access 12, 14324–14333. http://dx.doi.org/10.1109/ACCESS.2024.3356185.

Hsu, W.N., Bolte, B., Tsai, Y.H.H., Lakhotia, K., Salakhutdinov, R., Mohamed, A., 2021. HuBERT: Self-supervised speech representation learning by masked prediction of hidden units. IEEE/ACM Trans. Audio, Speech, Lang. Process. 29, 3451–3460. http://dx.doi.org/10.1109/TASLP.2021.3122291.

Hu, J., Liu, Y., Zhao, J., Jin, Q., 2021. MMGCN: Multimodal fusion via deep graph convolution network for emotion recognition in conversation. In: Zong, C., Xia, F., Li, W., Navigli, R. (Eds.), Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers). Association for Computational Linguistics, Online, pp. 5666–5675. http://dx.doi.org/10.18653/v1/2021.acl-long.440, URL: https://aclanthology.org/2021.acl-long.440/.

Ibrahim, H., Loo, C.K., Alnajjar, F., 2021. Speech emotion recognition by late fusion for bidirectional reservoir computing with random projection. IEEE Access 9, 122855–122871. http://dx.doi.org/10.1109/ACCESS.2021.3107858.

Jackson, P., Haq, S., 2014. Surrey Audio-Visual Expressed Emotion (Savee) Database. University of Surrey, Guildford, UK.

Jiang, Y., Li, W., Hossain, M.S., Chen, M., Alelaiwi, A., Al-Hammadi, M., 2020. A snapshot research and implementation of multimodal information fusion for data-driven emotion recognition. Inf. Fusion 53, 209–221. http://dx.doi.org/10.1016/j.inffus.2019.06.019.

Jiao, T., Guo, C., Feng, X., Chen, Y., Song, J., 2024. A comprehensive survey on deep learning multi-modal fusion: Methods, technologies and applications.. Comput. Mater. Contin. 80 (1), http://dx.doi.org/10.32604/cmc.2024.053204.

Kakuba, S., Poulose, A., Han, D.S., 2023. Deep learning approaches for bimodal speech emotion recognition: Advancements, challenges, and a multi-learning model. IEEE Access 11, 113769–113789. http://dx.doi.org/10.1109/ACCESS.2023.3325037.

Khan, Q.W., Ahmad, R., Rizwan, A., Khan, A.N., Park, C.W., Kim, D., 2024b. Multimodal fusion approaches for tourism: A comprehensive survey of data-sets, fusion techniques, recent architectures, and future directions. Comput. Electr. Eng. 116, 109220. http://dx.doi.org/10.1016/j.compeleceng.2024.109220.

Khan, M., Gueaieb, W., El Saddik, A., Kwon, S., 2024a. MSER: Multimodal speech emotion recognition using cross-attention with deep fusion. Expert Syst. Appl. 245, 122946. http://dx.doi.org/10.1016/j.eswa.2023.122946, URL: https://www.sciencedirect.com/science/article/pii/S0957417423034486.

Khare, S.K., Blanes-Vidal, V., Nadimi, E.S., Acharya, U.R., 2024. Emotion recognition and artificial intelligence: A systematic review (2014–2023) and research recommendations. Inf. Fusion 102, 102019. http://dx.doi.org/10.1016/j.inffus.2023.102019, URL: https://www.sciencedirect.com/science/article/pii/S1566253523003354.

Khatua, A., Khatua, A., Cambria, E., 2019. A tale of two epidemics: Contextual Word2Vec for classifying twitter streams during outbreaks. Inf. Process. Manage. 56 (1), 247–257. http://dx.doi.org/10.1016/j.ipm.2018.10.010.

Khurana, Y., Gupta, S., Sathyaraj, R., Raja, S.P., 2024. RobinNet: A multimodal speech emotion recognition system with speaker recognition for social interactions. IEEE Trans. Comput. Soc. Syst. 11 (1), 478–487. http://dx.doi.org/10.1109/TCSS.2022.3228649.

Kolbæk, M., Tan, Z.H., Jensen, S.H., Jensen, J., 2020. On loss functions for supervised monaural time-domain speech enhancement. IEEE/ACM Trans. Audio, Speech, Lang. Process. 28, 825–838. http://dx.doi.org/10.1109/TASLP.2020.2968738.

Kwon, S., et al., 2021. Att-net: Enhanced emotion recognition system using lightweight self-attention module. Appl. Soft Comput. 102, 107101. http://dx.doi.org/10.1016/j.asoc.2021.107101.

Latif, S., Rana, R., Khalifa, S., Jurdak, R., Qadir, J., Schuller, B., 2023. Survey of deep representation learning for speech emotion recognition. IEEE Trans. Affect. Comput. 14 (2), 1634–1654. http://dx.doi.org/10.1109/TAFFC.2021.3114365.

Le, Q.B., Tuan Trinh, K., Hung Son, N.D., Tran, P.N., Nguyen, C.T., Ngoc Minh Dang, D., 2024. MERSA: Multimodal emotion recognition with self-align embedding. In: 2024 International Conference on Information Networking. ICOIN, pp. 500–505. http://dx.doi.org/10.1109/ICOIN59985.2024.10572116.

Li, J., Cheng, K., Wang, S., Morstatter, F., Trevino, R.P., Tang, J., Liu, H., 2017. Feature selection: A data perspective. ACM Comput. Surv. 50 (6), http://dx.doi.org/10.1145/3136625.

Li, G., Hou, J., Liu, Y., Wei, J., 2023b. MPAF-CNN: Multiperspective aware and fine-grained fusion strategy for speech emotion recognition. Appl. Acoust. 214, 109658. http://dx.doi.org/10.1016/j.apacoust.2023.109658.

Li, M., Liu, X., Zhang, Y., Liang, W., 2024a. Late fusion multiview clustering via min-max optimization. IEEE Trans. Neural Networks Learn. Syst. 35 (7), 9417–9427. http://dx.doi.org/10.1109/TNNLS.2022.3233179.

Li, F., Luo, J., Liu, W., 2023a. Speech emotion recognition using multi-modal feature fusion network. In: 2023 IEEE 6th International Conference on Pattern Recognition and Artificial Intelligence. PRAI, pp. 884–888. http://dx.doi.org/10.1109/PRAI59366.2023.10332053.

Li, J., Wang, X., Lv, G., Zeng, Z., 2023c. GraphMFT: A graph network based multimodal fusion technique for emotion recognition in conversation. Neurocomputing 550, 126427. http://dx.doi.org/10.1016/j.neucom.2023.126427.

Li, S., Zhang, T., Chen, C.L.P., 2024b. SIA-net: Sparse interactive attention network for multimodal emotion recognition. IEEE Trans. Comput. Soc. Syst. 11 (5), 6782–6794. http://dx.doi.org/10.1109/TCSS.2024.3409715.

Lian, Z., Liu, B., Tao, J., 2021. CTNet: Conversational transformer network for emotion recognition. IEEE/ACM Trans. Audio, Speech, Lang. Process. 29, 985–1000. http://dx.doi.org/10.1109/TASLP.2021.3049898.

Liang, T., Glossner, J., Wang, L., Shi, S., Zhang, X., 2021. Pruning and quantization for deep neural network acceleration: A survey. Neurocomputing 461, 370–403. http://dx.doi.org/10.1016/j.neucom.2021.07.045, URL: https://www.sciencedirect.com/science/article/pii/S0925231221010894.

Lin, H., Zhang, P., Ling, J., Yang, Z., Lee, L.K., Liu, W., 2023. PS-mixer: A polar-vector and strength-vector mixer model for multimodal sentiment analysis. Inf. Process. Manage. 60 (2), 103229. http://dx.doi.org/10.1016/j.ipm.2022.103229, URL: https://www.sciencedirect.com/science/article/pii/S0306457322003302.

Liu, Y., Chen, X., Song, Y., Li, Y., Wang, S., Yuan, W., Li, Y., Zhao, Z., 2024b. Discriminative feature learning based on multi-view attention network with diffusion joint loss for speech emotion recognition. Eng. Appl. Artif. Intell. 137, 109219. http://dx.doi.org/10.1016/j.engappai.2024.109219.

Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., Stoyanov, V., 2019. Roberta: A robustly optimized bert pretraining approach. arXiv preprint arXiv:1907.11692.

Liu, Y., Sun, H., Guan, W., Xia, Y., Zhao, Z., 2022b. Multi-modal speech emotion recognition using self-attention mechanism and multi-scale fusion framework. Speech Commun. 139, 1–9. http://dx.doi.org/10.1016/j.specom.2022.02.006, URL: https://www.sciencedirect.com/science/article/pii/S0167639322000309.

Liu, J., Wang, H., Sun, M., Wei, Y., 2022a. Graph based emotion recognition with attention pooling for variable-length utterances. Neurocomputing 496, 46–55. http://dx.doi.org/10.1016/j.neucom.2022.05.007.

Liu, R., Zuo, H., Lian, Z., Schuller, B.W., Li, H., 2024a. Contrastive learning based modality-invariant feature acquisition for robust multimodal emotion recognition with missing modalities. IEEE Trans. Affect. Comput. 15 (4), 1856–1873. http://dx.doi.org/10.1109/TAFFC.2024.3378570.

Livingstone, S.R., Russo, F.A., 2018. The ryerson audio-visual database of emotional speech and song (RAVDESS): A dynamic, multimodal set of facial and vocal expressions in north American english. PLoS One 13 (5), e0196391. http://dx.doi.org/10.1371/journal.pone.0196391.

Lo, L., Hung, N., Lin, M., 2018. Angry versus furious: a comparison between valence and arousal in dimensional models of emotions. In: Emotions and their Influence on Our Personal, Interpersonal and Social Experiences. Routledge, pp. 75–86.

Lovisotto, G., Finnie, N., Munoz, M., Mummadi, C.K., Metzen, J.H., 2022. Give me your attention: Dot-product attention considered harmful for adversarial patch robustness. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 15234–15243. http://dx.doi.org/10.48550/arXiv.1710.10903.

Lu, S., Lu, J., An, K., Wang, X., He, Q., 2023. Edge computing on IoT for machine signal processing and fault diagnosis: A review. IEEE Internet Things J. 10 (13), 11093–11116. http://dx.doi.org/10.1109/JIOT.2023.3239944.

Luong, M.T., 2015. Effective approaches to attention-based neural machine translation. arXiv preprint arXiv:1508.04025.

Ma, X., Wu, Z., Jia, J., Xu, M., Meng, H., Cai, L., 2017. Speech emotion recognition with emotion-pair based framework considering emotion distribution information in dimensional emotion space. In: Interspeech. pp. 1238–1242. http://dx.doi.org/10.21437/Interspeech.2017-619.

Ma, Y., Zhang, Y., Bachinski, M., Fjeld, M., 2024. Emotion-aware voice assistants: Design, implementation, and preliminary insights. In: Proceedings of the Eleventh International Symposium of Chinese CHI. CHCHI '23, Association for Computing Machinery, New York, NY, USA, pp. 527–532. http://dx.doi.org/10.1145/3629606.3629665.

Madanian, S., Chen, T., Adeleye, O., Templeton, J.M., Poellabauer, C., Parry, D., Schneider, S.L., 2023. Speech emotion recognition using machine learning — A systematic review. Intell. Syst. Appl. 20, 200266. http://dx.doi.org/10.1016/j.iswa.2023.200266, URL: https://www.sciencedirect.com/science/article/pii/S2667305323000911.

Manalu, H.V., Rifai, A.P., 2024. Detection of human emotions through facial expressions using hybrid convolutional neural network-recurrent neural network algorithm. Intell. Syst. Appl. 21, 200339. http://dx.doi.org/10.1016/j.iswa.2024.200339, URL: https://www.sciencedirect.com/science/article/pii/S2667305324000152.

Mehrabian, A., 1996. Pleasure-arousal-dominance: A general framework for describing and measuring individual differences in temperament. Curr. Psychol. 14, 261–292. http://dx.doi.org/10.1007/BF02686918.

Meng, T., Shou, Y., Ai, W., Du, J., Liu, H., Li, K., 2024. A multi-message passing framework based on heterogeneous graphs in conversational emotion recognition. Neurocomputing 569, 127109. http://dx.doi.org/10.1016/j.neucom.2023.127109.

Menon, L.T., Neduziak, L.C.R., Barddal, J.P., Koerich, A.L., de Souza Britto, A., 2025. Dynamic modality and view selection for emotion recognition: An experimental study on missing modality evaluation. In: Palaiahnakote, S., Schuckers, S., Ogier, J.M., Bhattacharya, P., Pal, U., Bhattacharya, S. (Eds.), Pattern Recognition. ICPR 2024 International Workshops and Challenges. Springer Nature Switzerland, Cham, pp. 151–166.

Mohamed, A., Lee, H.y., Borgholt, L., Havtorn, J.D., Edin, J., Igel, C., Kirchhoff, K., Li, S.W., Livescu, K., Maaløe, L., Sainath, T.N., Watanabe, S., 2022. Self-supervised speech representation learning: A review. IEEE J. Sel. Top. Signal Process. 16 (6), 1179–1210. http://dx.doi.org/10.1109/JSTSP.2022.3207050.

Mohammadi, S., Sinaei, S., Balador, A., Flammini, F., 2023. Secure and efficient federated learning by combining homomorphic encryption and gradient pruning in speech emotion recognition. In: Meng, W., Yan, Z., Piuri, V. (Eds.), Information Security Practice and Experience. Springer Nature Singapore, Singapore, pp. 1–16.

Morcos, A., Raghu, M., Bengio, S., 2018. Insights on representational similarity in neural networks with canonical correlation. Adv. Neural Inf. Process. Syst. 31.

Müller, M., 2007. Dynamic time warping. In: Information Retrieval for Music and Motion. Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 69–84. http://dx.doi.org/10.1007/978-3-540-74048-3_4.

N., K.D., Patil, A., 2020. Multimodal emotion recognition using cross-modal attention and 1D convolutional neural networks. In: Interspeech 2020. pp. 4243–4247. http://dx.doi.org/10.21437/Interspeech.2020-1190.

Naderi, N., Nasersharif, B., 2023. Cross corpus speech emotion recognition using transfer learning and attention-based fusion of Wav2Vec2 and prosody features. Knowl.-Based Syst. 277, 110814. http://dx.doi.org/10.1016/j.knosys.2023.110814.

Nguyen, N.M., Nguyen, T.T., Nguyen, H.H., Tran, P.N., Dang, D.N.M., 2024a. Voice-based age and gender recognition: A comparative study of LSTM, RezoNet and hybrid CNNs-BiLSTM architecture. In: 2024 15th International Conference on Information and Communication Technology Convergence. ICTC, pp. 191–196. http://dx.doi.org/10.1109/ICTC62082.2024.10827387.

Nguyen, T.M., Tran, P.N., Dang, D.N.M., 2024b. Enhancing speech emotion recognition through knowledge distillation. In: 2024 15th International Conference on Information and Communication Technology Convergence. ICTC, pp. 197–202. http://dx.doi.org/10.1109/ICTC62082.2024.10826904.

Othmani, A., Kadoch, D., Bentounes, K., Rejaibi, E., Alfred, R., Hadid, A., 2021. Towards robust deep neural networks for affect and depression recognition from speech. In: Del Bimbo, A., Cucchiara, R., Sclaroff, S., Farinella, G.M., Mei, T., Bertini, M., Escalante, H.J., Vezzani, R. (Eds.), Pattern Recognition. ICPR International Workshops and Challenges. Springer International Publishing, Cham, pp. 5–19.

Pan, B., Hirota, K., Jia, Z., Dai, Y., 2023. A review of multimodal emotion recognition from datasets, preprocessing, features, and fusion methods. Neurocomputing 126866. http://dx.doi.org/10.1016/j.neucom.2023.126866.

Peña, D., Aguilera, A., Dongo, I., Heredia, J., Cardinale, Y., 2023. A framework to evaluate fusion methods for multimodal emotion recognition. IEEE Access 11, 10218–10237. http://dx.doi.org/10.1109/ACCESS.2023.3240420.

Peng, J., Wu, T., Zhang, W., Cheng, F., Tan, S., Yi, F., Huang, Y., 2023. A fine-grained modal label-based multi-stage network for multimodal sentiment analysis. Expert Syst. Appl. 221 (C), http://dx.doi.org/10.1016/j.eswa.2023.119721.

Pennington, J., Socher, R., Manning, C.D., 2014. Glove: Global vectors for word representation. In: Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing. EMNLP, pp. 1532–1543.

Pentari, A., Kafentzis, G., Tsiknakis, M., 2024. Speech emotion recognition via graph-based representations. Sci. Rep. 14 (1), 4484. http://dx.doi.org/10.1038/s41598-024-52989-2.

Pham, N.T., Phan, L.T., Dang, D.N.M., Manavalan, B., 2023. SER-fuse: An emotion recognition application utilizing multi-modal, multi-lingual, and multi-feature fusion. In: Proceedings of the 12th International Symposium on Information and Communication Technology. SOICT '23, Association for Computing Machinery, New York, NY, USA, pp. 870–877. http://dx.doi.org/10.1145/3628797.3628887.

Poria, S., Cambria, E., Bajpai, R., Hussain, A., 2017. A review of affective computing: From unimodal analysis to multimodal fusion. Inf. Fusion 37, 98–125. http://dx.doi.org/10.1016/j.inffus.2017.02.003.

Poria, S., Hazarika, D., Majumder, N., Naik, G., Cambria, E., Mihalcea, R., 2018. Meld: A multimodal multi-party dataset for emotion recognition in conversations. http://dx.doi.org/10.48550/arXiv.1810.02508, arXiv preprint arXiv:1810.02508.

Priyasad, D., Fernando, T., Sridharan, S., Denman, S., Fookes, C., 2023. Dual memory fusion for multimodal speech emotion recognition. In: Proc. INTERSPEECH, 2023, pp. 4543–4547. http://dx.doi.org/10.21437/Interspeech.2023-1090.

Qian, K., Zhang, Y., Gao, H., Ni, J., Lai, C.I., Cox, D., Hasegawa-Johnson, M., Chang, S., 2022. ContentVec: An improved self-supervised speech representation by disentangling speakers. In: Chaudhuri, K., Jegelka, S., Song, L., Szepesvari, C., Niu, G., Sabato, S. (Eds.), Proceedings of the 39th International Conference on Machine Learning. In: Proceedings of Machine Learning Research, vol. 162, PMLR, pp. 18003–18017, URL: https://proceedings.mlr.press/v162/qian22b.html.

Santoso, J., Yamada, T., Ishizuka, K., Hashimoto, T., Makino, S., 2022. Speech emotion recognition based on self-attention weight correction for acoustic and text features. IEEE Access 10, 115732–115743. http://dx.doi.org/10.1109/ACCESS.2022.3219094.

Saon, G., Tüske, Z., Audhkhasi, K., Kingsbury, B., 2019. Sequence noise injected training for end-to-end speech recognition. In: ICASSP 2019 - 2019 IEEE International Conference on Acoustics, Speech and Signal Processing. ICASSP, pp. 6261–6265. http://dx.doi.org/10.1109/ICASSP.2019.8683706.

Scarselli, F., Gori, M., Tsoi, A.C., Hagenbuchner, M., Monfardini, G., 2009. The graph neural network model. IEEE Trans. Neural Netw. 20 (1), 61–80. http://dx.doi.org/10.1109/TNN.2008.2005605.

Schuller, B., 2013. Audio features. In: Intelligent Audio Analysis. Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 41–97. http://dx.doi.org/10.1007/978-3-642-36806-6_6.

Sengupta, A., Ye, Y., Wang, R., Liu, C., Roy, K., 2019. Going deeper in spiking neural networks: VGG and residual architectures. Front. Neurosci. 13 - 2019, http://dx.doi.org/10.3389/fnins.2019.00095, URL: https://www.frontiersin.org/journals/neuroscience/articles/10.3389/fnins.2019.00095.

Shah Fahad, M., Ranjan, A., Yadav, J., Deepak, A., 2021. A survey of speech emotion recognition in natural environment. Digit. Signal Process. 110, 102951. http://dx.doi.org/10.1016/j.dsp.2020.102951, URL: https://www.sciencedirect.com/science/article/pii/S1051200420302967.

Shahin, I., Alomari, O.A., Nassif, A.B., Afyouni, I., Hashem, I.A., Elnagar, A., 2023. An efficient feature selection method for arabic and english speech emotion recognition using grey wolf optimizer. Appl. Acoust. 205, 109279.

Shang, Y., Fu, T., 2024. Multimodal fusion: a study on speech-text emotion recognition with the integration of deep learning. Intell. Syst. Appl. 200436.

Shixin, P., Kai, C., Tian, T., Jingying, C., 2022. An autoencoder-based feature level fusion for speech emotion recognition. Digit. Commun. Networks.

Shome, D., Etemad, A., 2024. Speech emotion recognition with distilled prosodic and linguistic affect representations. In: ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing. ICASSP, pp. 11976–11980. http://dx.doi.org/10.1109/ICASSP48485.2024.10448505.

Slimi, A., Hafar, N., Zrigui, M., Nicolas, H., 2022. Multiple models fusion for multi-label classification in speech emotion recognition systems. Procedia Comput. Sci. 207, 2875–2882. http://dx.doi.org/10.1016/j.procs.2022.09.345.

Song, H., Kang, K., Wei, Y., Zhang, H., Zhang, L., 2024. Speech emotion recognition based on multi coattention acoustic feature fusion. In: 2024 Second International Conference on Data Science and Information System. ICDSIS, IEEE, pp. 1–4.

Song, Z., Wei, H., Bai, L., Yang, L., Jia, C., 2023. GraphAlign: Enhancing accurate feature alignment by graph matching for multi-modal 3D object detection. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. ICCV, pp. 3358–3369.

Sousa, H., Guimarães, N., Jorge, A., Campos, R., 2023. GPT struct me: Probing GPT models on narrative entity extraction. In: 2023 IEEE/WIC International Conference on Web Intelligence and Intelligent Agent Technology. WI-IAT, pp. 383–387. http://dx.doi.org/10.1109/WI-IAT59888.2023.00063.

Sun, H., Dhingra, B., Zaheer, M., Mazaitis, K., Salakhutdinov, R., Cohen, W., 2018. Open domain question answering using early fusion of knowledge bases and text. In: Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing. pp. 4231–4242.

Tang, Y., Hu, Y., He, L., Huang, H., 2022. A bimodal network based on audio–text-interactional-attention with ArcFace loss for speech emotion recognition. Speech Commun. 143, 21–32. http://dx.doi.org/10.1016/j.specom.2022.07.004.

Targ, S., Almeida, D., Lyman, K., 2016. Resnet in resnet: Generalizing residual architectures. arXiv preprint arXiv:1603.08029.

Thanh, P.V., Huyen, N.T.T., Quan, P.N., Trang, N.T.T., 2024. A robust pitch-fusion model for speech emotion recognition in tonal languages. In: ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing. ICASSP, IEEE, pp. 12386–12390.

Tomar, P.S., Mathur, K., Suman, U., 2023. Unimodal approaches for emotion recognition: A systematic review. Cogn. Syst. Res. 77, 94–109. http://dx.doi.org/10.1016/j.cogsys.2022.10.012.

Tran, P.N., Vu, T.D.T., Dang, D.N.M., Pham, N.T., Tran, A.K., 2023a. Multi-modal speech emotion recognition: Improving accuracy through fusion of vggish and BERT features with multi-head attention. In: Vo, N.S., Tran, H.A. (Eds.), Industrial Networks and Intelligent Systems. Springer Nature Switzerland, Cham, pp. 148–158.

Tran, P.N., Vu, T.D.T., Truong Pham, N., Dang-Ngoc, H., Minh Dang, D.N., 2023b. Comparative analysis of multi-loss functions for enhanced multi-modal speech emotion recognition. In: 2023 14th International Conference on Information and Communication Technology Convergence. ICTC, pp. 425–429. http://dx.doi.org/10.1109/ICTC58733.2023.10392928.

Triantafyllopoulos, A., Schuller, B.W., 2021. The role of task and acoustic similarity in audio transfer learning: Insights from the speech emotion recognition case. In: ICASSP 2021 - 2021 IEEE International Conference on Acoustics, Speech and Signal Processing. ICASSP, pp. 7268–7272. http://dx.doi.org/10.1109/ICASSP39728.2021.9414896.

Tsai, Y.H.H., Bai, S., Liang, P.P., Kolter, J.Z., Morency, L.P., Salakhutdinov, R., 2019. Multimodal transformer for unaligned multimodal language sequences. In: Korhonen, A., Traum, D., Màrquez, L. (Eds.), Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics. Association for Computational Linguistics, Florence, Italy, pp. 6558–6569. http://dx.doi.org/10.18653/v1/P19-1656, URL: https://aclanthology.org/P19-1656/.

Tu, S., Dai, Q., Wu, Z., Cheng, Z.Q., Hu, H., Jiang, Y.G., 2023. Implicit temporal modeling with learnable alignment for video recognition. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. ICCV, pp. 19936–19947.

Tugwell, P., Tovey, D., 2021. PRISMA 2020. J. Clin. Epidemiol. 134, A5–A6. http://dx.doi.org/10.1016/j.jclinepi.2021.04.008, URL: https://www.sciencedirect.com/science/article/pii/S089543562100130X.

Umapathy, K., Krishnan, S., Rao, R.K., 2007. Audio signal feature extraction and classification using local discriminant bases. IEEE Trans. Audio, Speech, Lang. Process. 15 (4), 1236–1246. http://dx.doi.org/10.1109/TASL.2006.885921.

Vekkot, S., Gupta, D., 2022. Fusion of spectral and prosody modelling for multilingual speech emotion conversion. Knowl.-Based Syst. 242, 108360. http://dx.doi.org/10.1016/j.knosys.2022.108360, URL: https://www.sciencedirect.com/science/article/pii/S0950705122001356.

Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., Bengio, Y., 2018. Graph attention networks. In: International Conference on Learning Representations.

Vora, S., Lang, A.H., Helou, B., Beijbom, O., 2020. Pointpainting: Sequential fusion for 3d object detection. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 4604–4612.

Vyakaranam, A., Ramayah, B., Maul, T., 2024. Preliminary study: Speech emotion recognition in online teaching from the perspective of educators especially late deafened. In: 2024 2nd International Conference on Software Engineering and Information Technology. ICoSEIT, pp. 216–221. http://dx.doi.org/10.1109/ICoSEIT60086.2024.10497503.

Wang, Y., Guan, L., 2008. Recognizing human emotional state from audiovisual signals. IEEE Trans. Multimed. 10 (5), 936–946.

Wang, J., Guo, Z., Yang, C., Li, X., Cui, Z., 2023a. Multi-scale hybrid fusion network for mandarin audio-visual speech recognition. In: 2023 IEEE International Conference on Multimedia and Expo. ICME, IEEE, pp. 642–647.

Wang, Y., Li, Y., Cui, Z., 2023b. Incomplete multimodality-diffused emotion recognition. In: Oh, A., Naumann, T., Globerson, A., Saenko, K., Hardt, M., Levine, S. (Eds.), In: Advances in Neural Information Processing Systems, vol. 36, Curran Associates, Inc., pp. 17117–17128, URL: https://proceedings.neurips.cc/paper_files/paper/2023/file/372cb7805eaccb2b7eed641271a30eec-Paper-Conference.pdf.

Wang, P., Liu, A., Sun, X., 2025. Integrating emotion dynamics in mental health: A trimodal framework combining ecological momentary assessment, physiological measurements, and speech emotion recognition. Interdiscip. Med. n/a (n/a), e20240095. http://dx.doi.org/10.1002/INMD.20240095, URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/INMD.20240095, arXiv:https://onlinelibrary.wiley.com/doi/pdf/10.1002/INMD.20240095.

Wen, G., Liao, H., Li, H., Wen, P., Zhang, T., Gao, S., Wang, B., 2022. Self-labeling with feature transfer for speech emotion recognition. Knowl.-Based Syst. 254, 109589. http://dx.doi.org/10.1016/j.knosys.2022.109589.

Xie, J., Zhu, M., Hu, K., 2023. Fusion-based speech emotion classification using two-stage feature selection. Speech Commun. 152, 102955. http://dx.doi.org/10.1016/j.specom.2023.102955.

Xu, X., Li, D., Zhou, Y., Wang, Z., 2022. Multi-type features separating fusion learning for speech emotion recognition. Appl. Soft Comput. 130, 109648. http://dx.doi.org/10.1016/j.asoc.2022.109648.

Xu, M., Zhang, F., Zhang, W., 2021. Head fusion: Improving the accuracy and robustness of speech emotion recognition on the IEMOCAP and RAVDESS dataset. IEEE Access 9, 74539–74549. http://dx.doi.org/10.1109/ACCESS.2021.3067460.

Yan, P., Abdulkadir, A., Luley, P.P., Rosenthal, M., Schatte, G.A., Grewe, B.F., Stadelmann, T., 2024b. A comprehensive survey of deep transfer learning for anomaly detection in industrial time series: Methods, applications, and directions. IEEE Access 12, 3768–3789. http://dx.doi.org/10.1109/ACCESS.2023.3349132.

Yan, J., Li, H., Xu, F., Zhou, X., Liu, Y., Yang, Y., 2024a. Speech emotion recognition based on temporal-spatial learnable graph convolutional neural network. Electronics 13 (11), 2010.

Yao, Z., Wang, Z., Liu, W., Liu, Y., Pan, J., 2020. Speech emotion recognition using fusion of three multi-task learning-based classifiers: HSF-DNN, MS-CNN and LLD-RNN. Speech Commun. 120, 11–19. http://dx.doi.org/10.1016/j.specom.2020.03.005.

You, R., Guo, Z., Cui, L., Long, X., Bao, Y., Wen, S., 2020. Cross-modality attention with semantic graph embedding for multi-label classification. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 34, (07), pp. 12709–12716. http://dx.doi.org/10.1609/aaai.v34i07.6964.

Yu, L., Xu, F., Qu, Y., Zhou, K., 2024. Speech emotion recognition based on multi-dimensional feature extraction and multi-scale feature fusion. Appl. Acoust. 216, 109752. http://dx.doi.org/10.1016/j.apacoust.2023.109752.

Yuan, L., Huang, G., Li, F., Yuan, X., Pun, C.M., Zhong, G., 2023. RBA-GCN: Relational bilevel aggregation graph convolutional network for emotion recognition. IEEE/ACM Trans. Audio, Speech, Lang. Process. 31, 2325–2337. http://dx.doi.org/10.1109/TASLP.2023.3284509.

Zadeh, A., Cao, Y.S., Hessner, S., Liang, P.P., Poria, S., Morency, L.P., 2020. CMU-MOSEAS: A multimodal language dataset for spanish, portuguese, German and French. In: Proceedings of the Conference on Empirical Methods in Natural Language Processing. Conference on Empirical Methods in Natural Language Processing, vol. 2020, NIH Public Access, p. 1801.

Zadeh, A., Chen, M., Poria, S., Cambria, E., Morency, L.P., 2017. Tensor fusion network for multimodal sentiment analysis. In: Palmer, M., Hwa, R., Riedel, S. (Eds.), Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing. Association for Computational Linguistics, Copenhagen, Denmark, pp. 1103–1114. http://dx.doi.org/10.18653/v1/D17-1115, URL: https://aclanthology.org/D17-1115/.

Zaidi, S.A.M., Latif, S., Qadir, J., 2024. Enhancing cross-language multimodal emotion recognition with dual attention transformers. IEEE Open J. Comput. Soc. 5, 684–693. http://dx.doi.org/10.1109/OJCS.2024.3486904.

Zhang, Y., Cui, L., Sun, X., Xue, K., 2023b. Speech emotion recognition based on feature fusion and residual graph convolutional network. In: 2023 IEEE International Conference on Signal Processing, Communications and Computing. ICSPCC, pp. 1–6. http://dx.doi.org/10.1109/ICSPCC59353.2023.10400357.

Zhang, S., Li, C., 2022. Research on feature fusion speech emotion recognition technology for smart teaching. Mob. Inf. Syst. 2022 (1), 7785929. http://dx.doi.org/10.1155/2022/7785929.

Zhang, L.M., Ng, G.W., Leau, Y.B., Yan, H., 2023a. A parallel-model speech emotion recognition network based on feature clustering. IEEE Access 11, 71224–71234. http://dx.doi.org/10.1109/ACCESS.2023.3294274.

Zhang, X., Qu, S., Huang, J., Fang, B., Yu, P., 2018. Stock market prediction via multi-source multiple instance learning. IEEE Access 6, 50720–50728. http://dx.doi.org/10.1109/ACCESS.2018.2869735.

Zhang, Y., Sidibé, D., Morel, O., Mériaudeau, F., 2021b. Deep multimodal fusion for semantic image segmentation: A survey. Image Vis. Comput. 105, 104042.

Zhang, W., Yang, G., Zhang, N., Xu, L., Wang, X., Zhang, Y., Zhang, H., Del Ser, J., de Albuquerque, V.H.C., 2021a. Multi-task learning with multi-view weighted fusion attention for artery-specific calcification analysis. Inf. Fusion 71, 64–76. http://dx.doi.org/10.1016/j.inffus.2021.01.009, URL: https://www.sciencedirect.com/science/article/pii/S1566253521000154.

Zhao, J., Li, R., Jin, Q., 2021. Missing modality imagination network for emotion recognition with uncertain missing modalities. In: Zong, C., Xia, F., Li, W., Navigli, R. (Eds.), Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers). Association for Computational Linguistics, Online, pp. 2608–2618. http://dx.doi.org/10.18653/v1/2021.acl-long.203, URL: https://aclanthology.org/2021.acl-long.203/.

Zhao, Z., Zhao, Y., Bao, Z., Wang, H., Zhang, Z., Li, C., 2018. Deep spectrum feature representations for speech emotion recognition. In: Proceedings of the Joint Workshop of the 4th Workshop on Affective Social Multimedia Computing and First Multi-Modal Affective Computing of Large-Scale Multimedia Data. In: ASMMC-MMAC'18, Association for Computing Machinery, New York, NY, USA, pp. 27–33. http://dx.doi.org/10.1145/3267935.3267948.

Zheng, X., Du, Y., Qin, X., 2025. CoMaSa:Context multi-aware self-attention for emotional response generation. Neurocomputing 611, 128692. http://dx.doi.org/10.1016/j.neucom.2024.128692, URL: https://www.sciencedirect.com/science/article/pii/S0925231224014632.

Zheng, Q., Zhu, J., Li, Z., Pang, S., Wang, J., Li, Y., 2020. Feature concatenation multi-view subspace clustering. Neurocomputing 379, 89–102. http://dx.doi.org/10.1016/j.neucom.2019.10.074, URL: https://www.sciencedirect.com/science/article/pii/S0925231219315127.

Zhu, C., Sun, C., Liu, Y., Chen, J., Luo, K., 2025. Enhancing speech emotion recognition with speech dynamic modeling and multi-modal knowledge distillation. In: ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing. ICASSP, pp. 1–5. http://dx.doi.org/10.1109/ICASSP49660.2025.10889655.

Zou, H., Si, Y., Chen, C., Rajan, D., Chng, E.S., 2022. Speech emotion recognition with co-attention based multi-level acoustic information. In: ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing. ICASSP, IEEE, pp. 7367–7371.

Zuo, H., Liu, R., Zhao, J., Gao, G., Li, H., 2023. Exploiting modality-invariant feature for robust multimodal emotion recognition with missing modalities. In: ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing. ICASSP, pp. 1–5. http://dx.doi.org/10.1109/ICASSP49357.2023.10095836.